Selective Inattention to Interest Rates

Pierfrancesco Mei
Goldman Sachs
pierfrancescomei.com

Tim de Silva
Stanford GSB
timdesilva.me

April 2026

Motivation

- Households' macro expectations suggest they are very uninformed on **average**
  - Level of expectations is often systematically biased Weber et al. 22
  - Substantial dispersion in expectations across people Mankiw et al. 04
  - Errors in their expectations are predictable ex-ante Bordalo et al. 20
- Motivated adding information frictions to quantitative macro models Auclert et al. 20
  - ⇒ **Average** expectation is slow-moving and under-reacts Coibion-Gorodnichenko 12, 15
  - ⇒ Aggregate responses to shocks are "hump-shaped", like in data Christiano et al. 05

Motivation

- Households' macro expectations suggest they are very uninformed on **average**
  - Level of expectations is often systematically biased Weber et al. 22
  - Substantial dispersion in expectations across people Mankiw et al. 04
  - Errors in their expectations are predictable ex-ante Bordalo et al. 20
- Motivated adding information frictions to quantitative macro models Auclert et al. 20
- Introspection: macro expectations much more important for "**big**" decisions
  - These **big** decisions also tend to occur less frequently
  - Example: interest rates important when **buying a house**, but less so for groceries

Motivation

- Households' macro expectations suggest they are very uninformed on **average**
  - Level of expectations is often systematically biased Weber et al. 22
  - Substantial dispersion in expectations across people Mankiw et al. 04
  - Errors in their expectations are predictable ex-ante Bordalo et al. 20
- Motivated adding information frictions to quantitative macro models Auclert et al. 20
- Introspection: macro expectations much more important for "**big**" decisions
- **Question**: Do HHs **select** into paying attention based on **types** of decisions?

Motivation

- Households' macro expectations suggest they are very uninformed on **average**
  - Level of expectations is often systematically biased Weber et al. 22
  - Substantial dispersion in expectations across people Mankiw et al. 04
  - Errors in their expectations are predictable ex-ante Bordalo et al. 20
- Motivated adding information frictions to quantitative macro models Auclert et al. 20
- Introspection: macro expectations much more important for "**big**" decisions
- **Question**: Do HHs **select** into paying attention based on **types** of decisions?
- **If yes**, how do the macro implications of this **selection** differ?
  - Belief heterogeneity $\Rightarrow$ **average** may not be the relevant object Miller 77, Afrouzi et al. 24

THIS PAPER

Is there selective inattention to interest rates based on $\underbrace{\textbf{durables purchases}}_{\text{“decision-making” (DM)}}$?

1. Use existing surveys to study how interest rate expectations differ based on DM
   - Benefit: high-quality data on expectations
   - Cost: imprecise identification of **DM status** + hard to isolate attention

This Paper

Is there selective inattention to interest rates based on $\underbrace{\textbf{durables purchases}}_{\text{“decision-making” (DM)}}$?

1. Use existing surveys to study how interest rate expectations differ based on DM
2. Conduct a new survey to identify how macro attention changes based on DM
   - Benefit: better identify **DM status** + elicit information acquisition directly
   - Cost: hard to study accuracy in expectations given one cross-section

THIS PAPER

~~Is there selective inattention to interest rates based on durables purchases?~~ ✓

How does selective inattention affect **aggregate responses** to interest rates?

1. Use existing surveys to study how interest rate expectations differ based on DM
2. Conduct a new survey to identify how macro attention changes based on DM
3. Develop incomplete markets model with $\overbrace{\text{durables}}^{\text{DM in model}}$ + rational inattention to rates
   - Use patterns in IA from survey to discipline information cost parameter
   - Compare **model IRFs** to level and volatility of rates with $\underbrace{\text{exogenous inattention}}_{\text{DM} \perp \text{beliefs} \Rightarrow \text{no selection}}$

Preview of Key Takeaways

- Data: Households are selectively inattentive to interest rates
  - IA is concentrated around durables purchases, where beliefs are more accurate
- Model: **Like** exogenous inattention, selective inattention generates:
  1. Slow-moving aggregate beliefs
  2. Dampened responses of non-durable consumption to interest rates

Preview of Key Takeaways

- Data: Households are selectively inattentive to interest rates
  - IA is concentrated around durables purchases, where beliefs are more accurate
- Model: **Like** exogenous inattention, selective inattention generates:
  1. Slow-moving aggregate beliefs
  2. Dampened responses of non-durable consumption to interest rates
- Model: **Unlike** exogenous inattention, selective inattention:
  1. Shifts the **composition** of spending responses to rate cuts towards durables
  2. **Accelerates** the impact of larger rate cuts
  3. **Dampens** responses to changes in volatility closer to the data

Preview of Key Takeaways

- Data: Households are selectively inattentive to interest rates
  - IA is concentrated around durables purchases, where beliefs are more accurate
- Model: **Like** exogenous inattention, selective inattention generates:
  1. Slow-moving aggregate beliefs
  2. Dampened responses of non-durable consumption to interest rates
- Model: **Unlike** exogenous inattention, selective inattention:
  1. Shifts the **composition** of spending responses to rate cuts towards durables
  2. **Accelerates** the impact of larger rate cuts
  3. **Dampens** responses to changes in volatility closer to the data

Takeaway: Aggregate inattention hides substantial selection into attention that can be measured, modeled, and has different implications!

Literature

Related Literature and Contribution

- Household expectation formation $\rightarrow$ *aggregate inattention masks* ***selection***

  Coibion & Gorodnichenko (2012, 2015), Bordalo et al. (2020), Weber et al. (2022), ...

Related Literature and Contribution

- Household expectation formation $\rightarrow$ *aggregate inattention masks* ***selection***
- Durables adjustment $+$ rational inattention $\rightarrow$ *first model* ***combining*** *the two*

  Berger & Vavra (2015), McKay & Wieland (2021), Beraja & Wolf (2022), Beraja & Zorzi (2024), Sims (2003), Mackowiak et al. (2023), Alvarez et al. (2012), Afrouzi et al. (2024)

Related Literature and Contribution

- Household expectation formation $\rightarrow$ *aggregate inattention masks* ***selection***
- Durables adjustment $+$ rational inattention $\rightarrow$ *first model* ***combining*** *the two*
- Macro models with endogenous household inattention $\rightarrow$ *inattention determined by* ***type of decision*** *that households make*

  Reis (2006), Broer et al. (2022), Guerreiro (2023), Ahn et al. (2024)

Related Literature and Contribution

- Household expectation formation $\rightarrow$ *aggregate inattention masks* ***selection***
- Durables adjustment $+$ rational inattention $\rightarrow$ *first model* ***combining*** *the two*
- Macro models with endogenous household inattention $\rightarrow$ *inattention determined by* ***type of decision*** *that households make*
- Eliciting information acquisition in the field $\rightarrow$ *use IA elicitations to* ***identify*** *structural models*

  Coibion et al. (2018), Roth et al. (2022), Mikosch et al. (2024)

Related Literature and Contribution

- Household expectation formation $\rightarrow$ *aggregate inattention masks* ***selection***
- Durables adjustment $+$ rational inattention $\rightarrow$ *first model* ***combining*** *the two*
- Macro models with endogenous household inattention $\rightarrow$ *inattention determined by* ***type of decision*** *that households make*
- Eliciting information acquisition in the field $\rightarrow$ *use IA elicitations to* ***identify*** *structural models*
- Effects of (interest rate) uncertainty $\rightarrow$ *depends on* ***endogeneity of inattention***

Bloom (2014), Vavra (2014), Bloom et al. (2020), Cremers et al. (2021), Ilut et al. (2025)

Outline

1. Existing Surveys: Expectations Accuracy around Decision-Making
2. New Survey: Information Acquisition around Decision-Making
3. Incomplete Markets Model with Selective Inattention
4. Interest Rate Passthrough with Selective Inattention
5. Conclusion

Outline

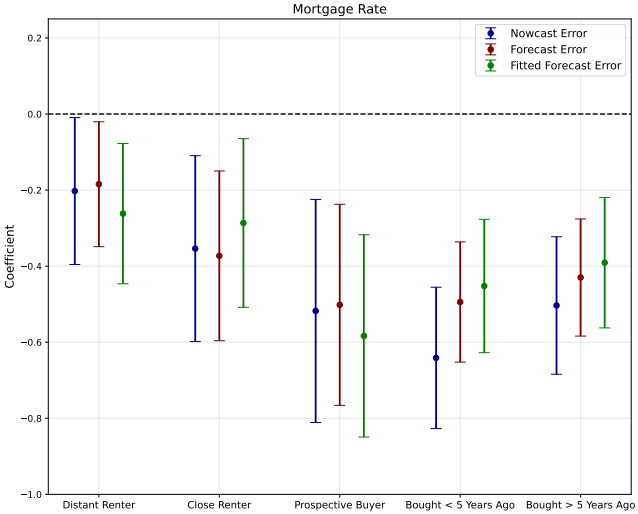
1. Existing Surveys: Expectations Accuracy around Decision-Making
2. New Survey: Information Acquisition around Decision-Making
3. Incomplete Markets Model with Selective Inattention
4. Interest Rate Passthrough with Selective Inattention
5. Conclusion

Survey 1/2: NY Fed Survey of Consumer Expectations

- **Sample**: repeated cross-section of $\sim$ 8K respondents in 2013-2023
- Variables of interest:
  1. **Nowcasts** of current average 30-year fixed mortgage rate
  2. **Forecasts** of one-year ahead mortgage rate and inflation
  3. **DM status** based on distance from past or (intended) future home purchase
- Construct errors using 30-year fixed rate in Freddie Mac PMMS and CPI
- Run the following regression:

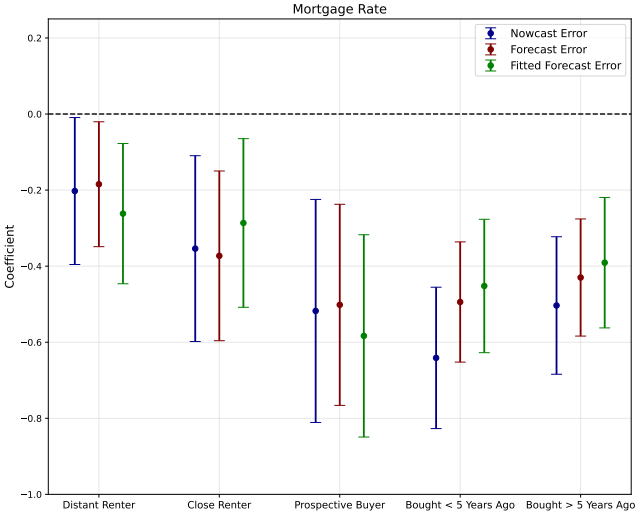
$$\log |\text{Error}_{it}| = \sum_{s} \beta_s \cdot \mathbf{1}\left(\text{DM Status}_{it} = s\right) + \text{Controls}_{it} + \delta_t + \epsilon_{it}$$

Decision-Makers Have More Accurate Beliefs



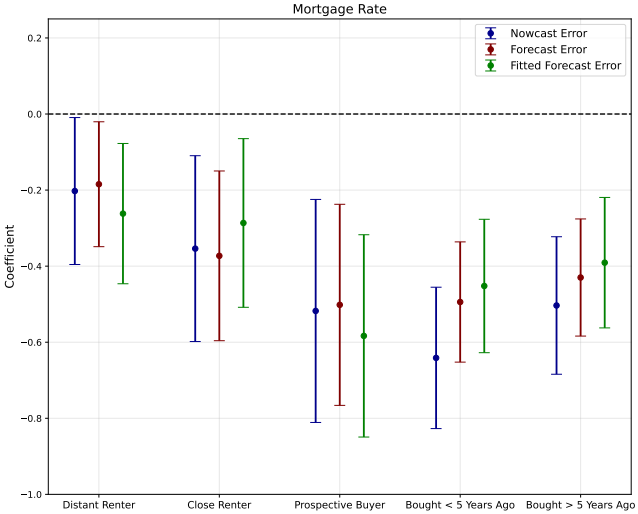
Errors of prospective buyers ≈ **50% lower** than those with no purchase plan

Decision-Makers Have More Accurate Beliefs



≈ **2x** difference between individuals in top and bottom terciles of income or education

Decision-Makers Have More Accurate Beliefs



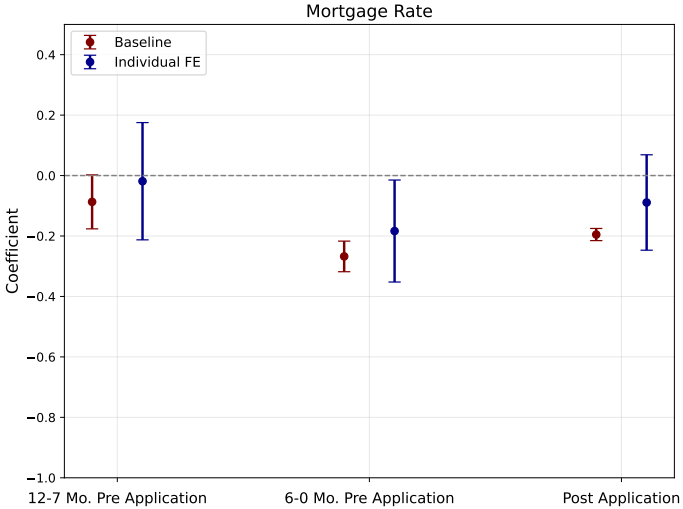
Almost all of forecasting gain comes from nowcasting improvement

Survey 2/2: ECB Survey of Consumer Expectations

- **Sample**: **panel** of $\sim$ 130K individuals in 2020-2024 from six largest countries
  - Restriction: only include individuals who rented at some point in survey
  - Note: quarterly frequency $\Rightarrow$ short panel
- Variables of interest:
  1. **Forecasts** of one-year ahead mortgage rates, inflation, GDP, and unemployment
  2. **DM status** based on distance from mortgage application
- Construct errors based on country-specific realizations
- Run the following regression:

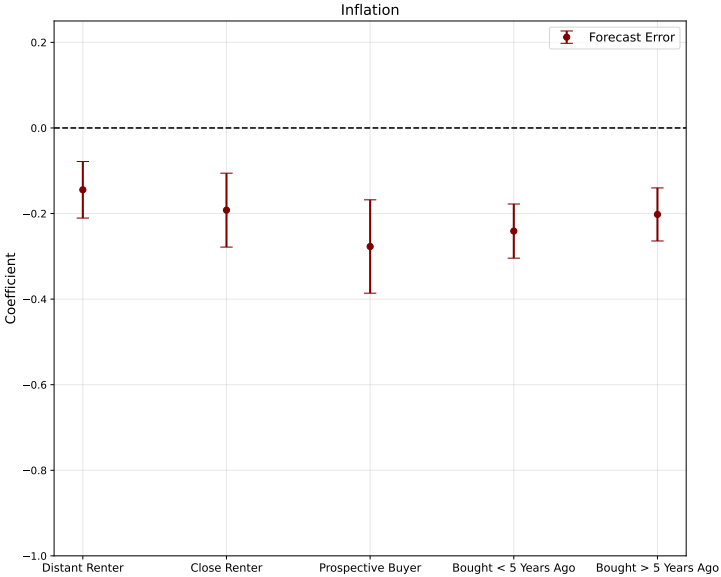
$$\log |\text{Error}_{it}| = \sum_s \beta_s \cdot \mathbf{1}\left(\text{DM Status}_{it} = s\right) + \text{Controls}_{it} + \textbf{Tenure}_{it} + \delta_{ct} + \lambda_{\mathbf{i}} + \epsilon_{it}$$

Decision-Makers Have More Accurate Beliefs: ECB

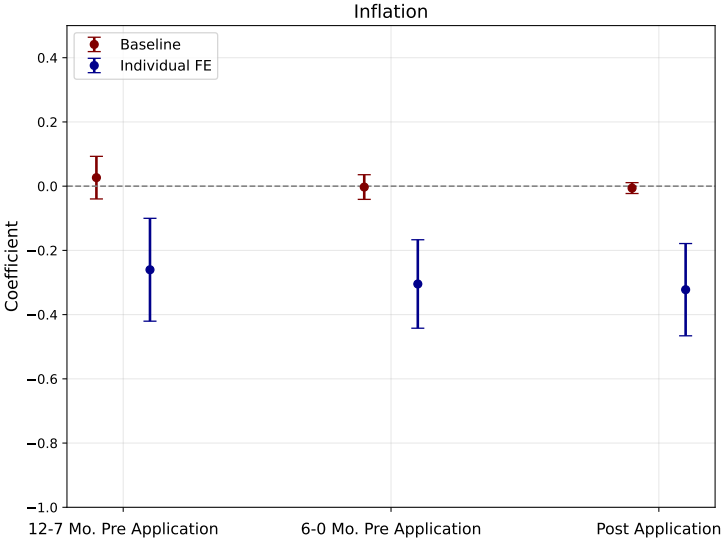


Mortgage rate forecast errors fall by about **25%** pre-application

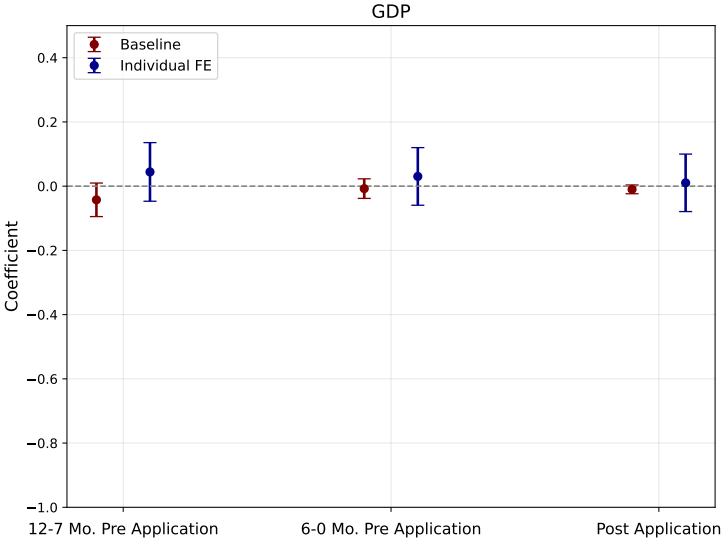
Smaller Differences for Other Variables: Inflation in NY Fed



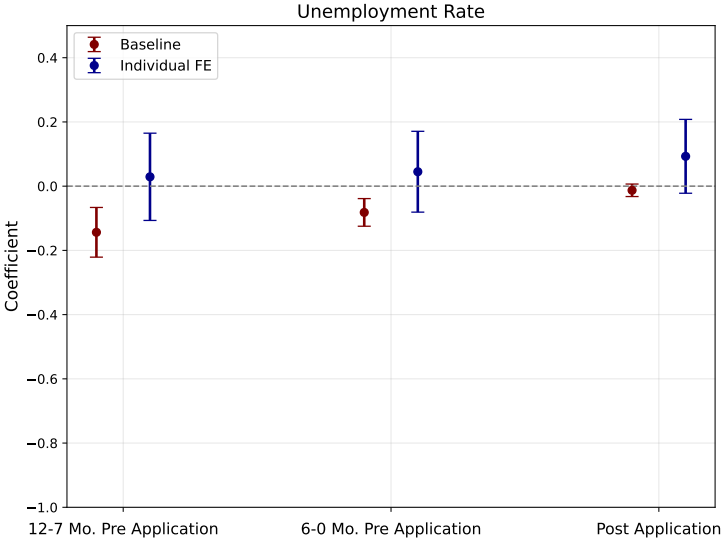
Smaller Differences for Other Variables: Inflation in ECB



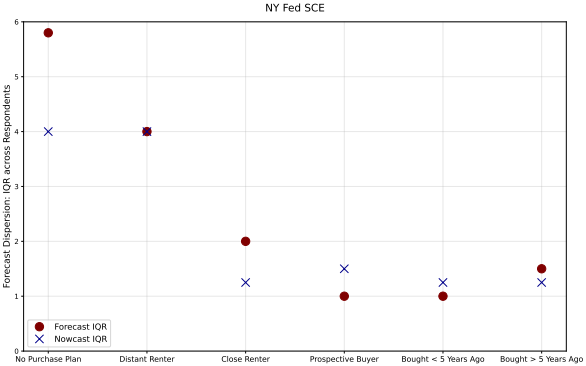
Smaller Differences for Other Variables: GDP in ECB

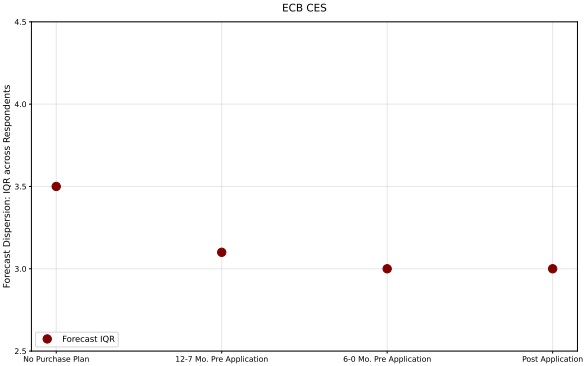


Smaller Differences for Other Variables: UR in ECB



Less Dispersion among DMs' Beliefs: Both Surveys





Outline

1. Existing Surveys: Expectations Accuracy around Decision-Making
2. New Survey: Information Acquisition around Decision-Making
3. Incomplete Markets Model with Selective Inattention
4. Interest Rate Passthrough with Selective Inattention
5. Conclusion

Survey Description

We design and conduct a cross-sectional survey of U.S. households via Prolific

Survey Description

We design and conduct a cross-sectional survey of U.S. households via Prolific

Survey Innovations

- Direct measure of distance from durable adjustments
  - Primary home purchase
  - Car purchases
- Measures of information acquisition other than forecasting performance
  - Last active search for information about key variables

Survey Description

We design and conduct a cross-sectional survey of U.S. households via Prolific

Survey Innovations

- Direct measure of distance from durable adjustments
  - Primary home purchase
  - Car purchases
- Measures of information acquisition other than forecasting performance
  - Last active search for information about key variables

Main Blocks

1. Home decision-making: distance from primary home purchase

Survey Description

We design and conduct a cross-sectional survey of U.S. households via Prolific

Survey Innovations

- Direct measure of distance from durable adjustments
  - Primary home purchase
  - Car purchases
- Measures of information acquisition other than forecasting performance
  - Last active search for information about key variables

Main Blocks

1. Home decision-making
2. Other decisions: distance from car purchase + other major financial decisions

Survey Description

We design and conduct a cross-sectional survey of U.S. households via Prolific

Survey Innovations

- Direct measure of distance from durable adjustments
  - Primary home purchase
  - Car purchases
- Measures of information acquisition other than forecasting performance
  - Last active search for information about key variables

Main Blocks

1. Home decision-making
2. Other decisions
3. Information acquisition: time since last search + type/source of info searched

Survey Description

We design and conduct a cross-sectional survey of U.S. households via Prolific

Survey Innovations

- Direct measure of distance from durable adjustments
  - Primary home purchase
  - Car purchases
- Measures of information acquisition other than forecasting performance
  - Last active search for information about key variables

Main Blocks

1. Home decision-making
2. Other decisions
3. Information acquisition
4. Macro expectations: beliefs about mortgage rates, T-Bill rates, and inflation

Survey Description

We design and conduct a cross-sectional survey of U.S. households via Prolific

Survey Innovations

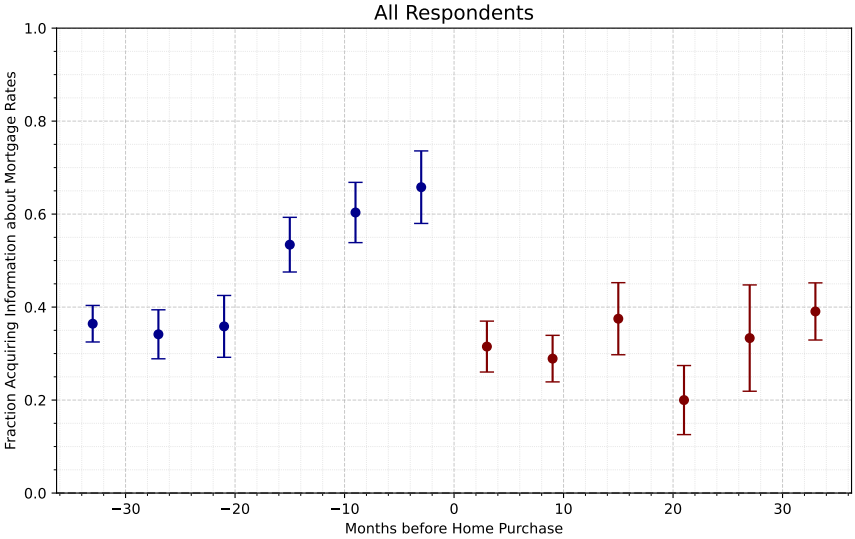
- Direct measure of distance from durable adjustments
  - Primary home purchase
  - Car purchases
- Measures of information acquisition other than forecasting performance
  - Last active search for information about key variables

Main Blocks

1. Home decision-making
2. Other decisions
3. Information acquisition
4. Macro expectations
5. Background & financial situation: info on household's balance-sheet using SCF format, demographics, job relocations

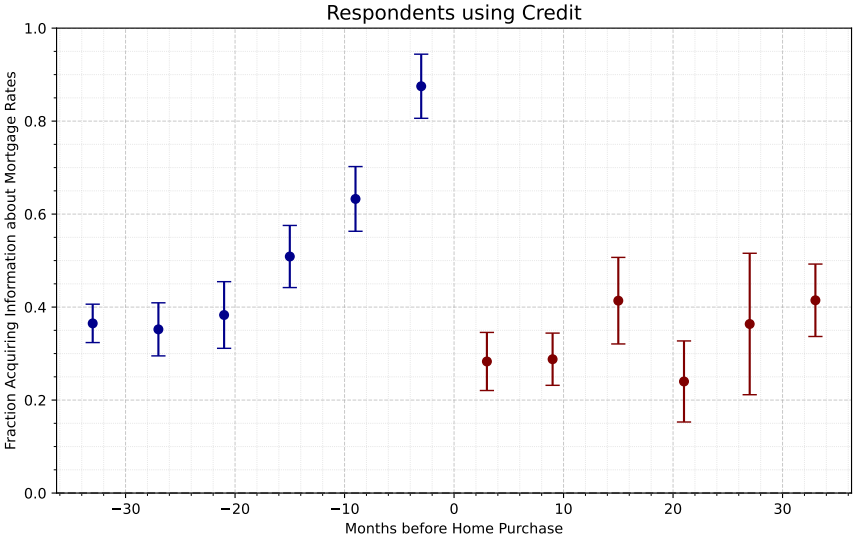
Questions

Information Acquisition is Concentrated Pre-Decision



Sources | Heterogeneity | Owners | Values

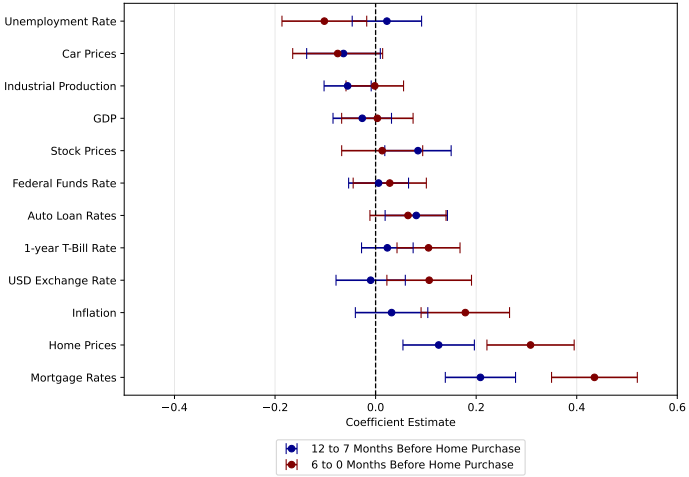
Information Acquisition is Concentrated Pre-Decision



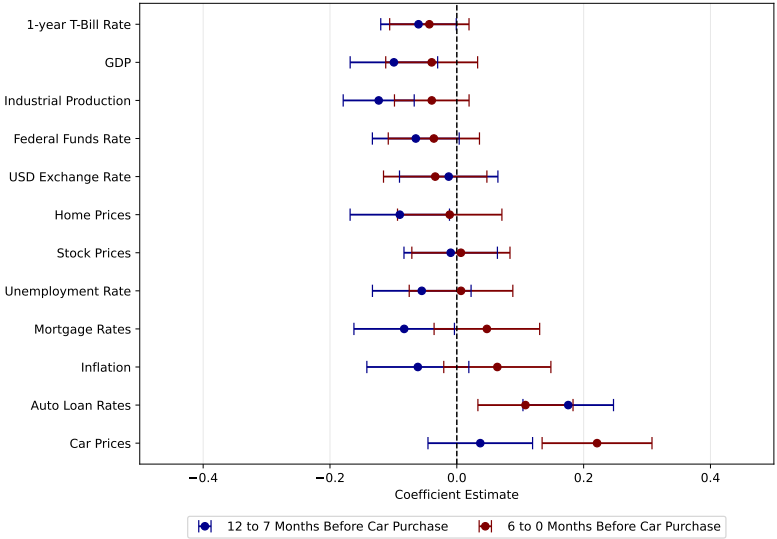
Sources | Heterogeneity | Owners | Values

IA is Concentrated on Decision-Relevant Variables

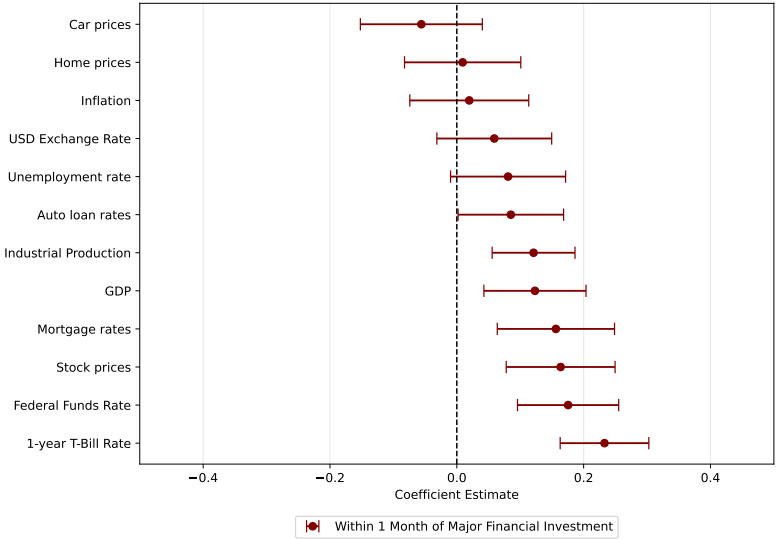
$$\text{Info. Acquisition}_i = \sum_d \beta_d \cdot \mathbf{1}\left(\text{Home Distance}_i = d\right) + \text{Controls}_i + \text{Other Distances}_i + \epsilon_i$$



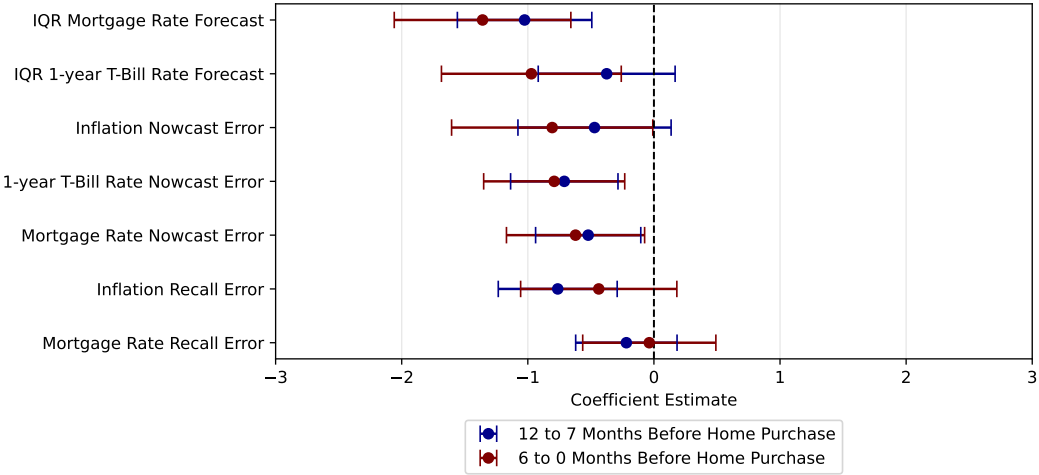
Patterns in Information Acquisition are Decision-Specific



Patterns in Information Acquisition are Decision-Specific



DMs' Beliefs Are (Weakly) More Informed and Less Dispersed



Additional Results from New Survey

- Information acquisition (IA) is primarily about the current values of variables
- Primary source of information is internet search
- No difference in IA based on mortgage type or refinancing plans
- Positive effect of decision-making due to job relocations on IA
- IA is directly associated with belief accuracy and uncertainty
- Similar improvement in belief accuracy for home purchases to SCE
- Reductions in subjective uncertainty for home purchases

Taking Stock: 5 Facts

1. Households close to **durables** purchases have more **accurate** macro expectations
2. Improvement in expectation accuracy is largest for **interest rates**
3. Decision-makers' beliefs are less **dispersed** and **uncertain**
4. Households concentrate **information acquisition** around durables purchases
5. Households' information acquisition is about **decision-relevant variables**

Taking Stock: 5 Facts

1. Households close to **durables** purchases have more **accurate** macro expectations
2. Improvement in expectation accuracy is largest for **interest rates**
3. Decision-makers' beliefs are less **dispersed** and **uncertain**
4. Households concentrate **information acquisition** around durables purchases
5. Households' information acquisition is about **decision-relevant variables**

Next: what are the macroeconomic implications of this **selective inattention**?

Outline

1. Existing Surveys: Expectations Accuracy around Decision-Making
2. New Survey: Information Acquisition around Decision-Making
3. Incomplete Markets Model with Selective Inattention
4. Interest Rate Passthrough with Selective Inattention
5. Conclusion

Model Overview

$$\underbrace{\overbrace{\text{Partial equilibrium incomplete markets model}}^{\text{Bewley-Huggett-Aiyagari}} + \text{durables}}_{\text{household block of McKay-Wieland 2021}} + \underbrace{\text{dynamic info. acquisition}}_{\text{rational inattention}}$$

Model Overview

Partial equilibrium incomplete markets model + durables + dynamic info. acquisition

Decision-Making

Given beliefs, HHs choose non-durables $\mathbf{c}$
and durables $\mathbf{d}'$ subject to:

- Income risk + collateralized borrowing
- Stochastic interest rate r
- Depreciation of durables stock
- Durables adjustment costs
- Operating + maintenance costs
- Match-quality shocks (e.g. job change)

Model Overview

Partial equilibrium incomplete markets model + durables + dynamic info. acquisition

Decision-Making

Given beliefs, HHs choose non-durables $\mathbf{c}$ and durables $\mathbf{d}'$ subject to:

- Income risk + collateralized borrowing
- Stochastic interest rate r
- Depreciation of durables stock
- Durables adjustment costs
- Operating + maintenance costs
- Match-quality shocks (e.g. job change)

$$\Downarrow$$

Rich model of how beliefs about $r \longrightarrow \mathbf{c}, \mathbf{d}'$

Model Overview

Partial equilibrium incomplete markets model + durables + dynamic info. acquisition

Decision-Making

Given beliefs, HHs choose non-durables $\mathbf{c}$ and durables $\mathbf{d}'$ subject to:

- Income risk + collateralized borrowing
- Stochastic interest rate r
- Depreciation of durables stock
- Durables adjustment costs
- Operating + maintenance costs
- Match-quality shocks (e.g. job change)

$$\Downarrow$$

Rich model of how beliefs about $r \longrightarrow \mathbf{c}, \mathbf{d}'$

Information Acquisition

HHs receive signals of endogenous precision about current r = rational inattention

- Cost of signals = $\omega \times$ mutual info.
- Benefit of signals = better choice of $\mathbf{c}, \mathbf{d}'$
- Interest rate is persistent $\Rightarrow$ prior beliefs are state variables

Model Overview

Partial equilibrium incomplete markets model + durables + dynamic info. acquisition

Decision-Making

Given beliefs, HHs choose non-durables $\mathbf{c}$ and durables $\mathbf{d}'$ subject to:

- Income risk + collateralized borrowing
- Stochastic interest rate r
- Depreciation of durables stock
- Durables adjustment costs
- Operating + maintenance costs
- Match-quality shocks (e.g. job change)

$\Downarrow$

Rich model of how beliefs about $r \longrightarrow \mathbf{c}, \mathbf{d}'$

Information Acquisition

HHs receive signals of endogenous precision about current r = rational inattention

- Cost of signals = $\omega \times$ mutual info.
- Benefit of signals = better choice of $\mathbf{c}, \mathbf{d}'$
- Interest rate is persistent $\Rightarrow$ prior beliefs are state variables

$\Downarrow$

Endogenous beliefs about r that come from dynamic information acquisition

Decision-Making Problem, Given Beliefs

- Households do not observe the **current** interest rate, r, but know DGP
- Define households' idiosyncratic information set by $\mathcal{I}$

Decision-Making Problem, Given Beliefs

- Households do not observe the **current** interest rate, r, but know DGP
- Define households' idiosyncratic information set by $\mathcal{I}$
- Given state $\mathbf{x} = (b, d, r, y, \xi, \mathcal{I})$, households solve:

Decision-Making Problem, Given Beliefs

- Households do not observe the **current** interest rate, r, but know DGP
- Define households' idiosyncratic information set by $\mathcal{I}$
- Given state $\mathbf{x} = (b, d, r, y, \xi, \mathcal{I})$, households solve:

$$\mathbf{c}(\mathbf{x}), \mathbf{d}'(\mathbf{x}) = \arg\max_{c, d'} U\Big(c, m(d')\Big) + \beta \cdot \mathbf{E} V\Big(\mathbf{x}' \mid \mathcal{I}\Big)$$

Decision-Making Problem, Given Beliefs

- Households do not observe the **current** interest rate, r, but know DGP
- Define households' idiosyncratic information set by $\mathcal{I}$
- Given state $\mathbf{x} = (b, d, r, y, \xi, \mathcal{I})$, households solve:

$$\mathbf{c}(\mathbf{x}), \mathbf{d}'(\mathbf{x}) = \arg\max_{c, d'} U\Big(c, m(d')\Big) + \beta \cdot \mathbf{E} V\Big(\mathbf{x}' \mid \mathcal{I}\Big)$$

$$c + b' + d' = y + \left[\exp(r) + \tau_b \mathbf{1}_{b<0}\right] b + (1 - \delta)\, d - A(d, d'),$$

Decision-Making Problem, Given Beliefs

- Households do not observe the **current** interest rate, r, but know DGP
- Define households' idiosyncratic information set by $\mathcal{I}$
- Given state $\mathbf{x} = (b, d, r, y, \xi, \mathcal{I})$, households solve:

$$\mathbf{c}(\mathbf{x}), \mathbf{d}'(\mathbf{x}) = \arg\max_{c, d'} U\Big(c, m(d')\Big) + \beta \cdot \mathbf{E}\, V\Big(\mathbf{x}' \mid \mathcal{I}\Big)$$

$$c + b' + d' = y + \left[\exp(r) + \tau_b \mathbf{1}_{b<0}\right] b + (1 - \delta)\, d - A(d, d'), \quad b' \geq -\lambda d'$$

Decision-Making Problem, Given Beliefs

- Households do not observe the **current** interest rate, r, but know DGP
- Define households' idiosyncratic information set by $\mathcal{I}$
- Given state $\mathbf{x} = (b, d, r, y, \xi, \mathcal{I})$, households solve:

$$\mathbf{c}(\mathbf{x}), \mathbf{d}'(\mathbf{x}) = \arg\max_{c,d'} U\Big(c, m(d')\Big) + \beta \cdot \mathbf{E}V\Big(\mathbf{x}' \mid \mathcal{I}\Big)$$

$$c + b' + d' = y + [\exp(r) + \tau_b \mathbf{1}_{b<0}]\, b + (1-\delta)\, d - \mathbf{A(d, d')}, \quad b' \geq -\lambda d'$$

$$\mathbf{A(d, d')} = \begin{cases} \underbrace{\nu \cdot d}_{\text{op. costs}} & \text{if } d' = \underbrace{(1-\delta)d}_{\text{depreciation}} + \underbrace{\delta \cdot \chi \cdot d}_{\text{maint. costs}} \\ \underbrace{\nu \cdot d}_{\text{op. costs}} + \underbrace{f \cdot (1-\delta)d}_{\text{fixed adj. cost}} & \text{else} \end{cases}$$

Decision-Making Problem, Given Beliefs

- Households do not observe the **current** interest rate, r, but know DGP
- Define households' idiosyncratic information set by $\mathcal{I}$
- Given state $\mathbf{x} = (b, d, r, y, \xi, \mathcal{I})$, households solve:

$$\mathbf{c}(\mathbf{x}), \mathbf{d}'(\mathbf{x}) = \arg\max_{c,d'} U\Big(c, \mathbf{m}(d')\Big) + \beta \cdot \mathbf{E}\, V\Big(\mathbf{x}' \mid \mathcal{I}\Big)$$

$$c + b' + d' = y + \left[\exp(r) + \tau_b \mathbf{1}_{b<0}\right] b + (1-\delta)\, d - A(d, d'), \quad b' \geq -\lambda d'$$

$$\mathbf{m}(d') = d' \times \max\left\{\xi, \mathbf{1}_{d' \neq (1-\delta+\delta\cdot\chi)d}\right\}, \quad \xi \sim \text{Bern}(\bar{\xi}) = \text{match-quality shock}$$

$\xi = 0 \Rightarrow$ have to adjust for **exogenous** reasons (e.g. job relocation)

Decision-Making Problem, Given Beliefs

- Households do not observe the **current** interest rate, r, but know DGP
- Define households' idiosyncratic information set by $\mathcal{I}$
- Given state $\mathbf{x} = (b, d, r, y, \xi, \mathcal{I})$, households solve:

$$\mathbf{c}(\mathbf{x}), \mathbf{d}'(\mathbf{x}) = \arg\max_{c, d'} U\Big(c, m(d')\Big) + \beta \cdot \mathbf{E} V\Big(\mathbf{x}' \mid \mathcal{I}\Big)$$

$$c + b' + d' = y + \left[\exp(r) + \tau_b \mathbf{1}_{b<0}\right] b + (1 - \delta)\, d - A(d, d'), \quad b' \geq -\lambda d'$$

- $\log y$ follows AR1 $+$ observed by households
- r follows an AR1 that households correctly perceive

Information Acquisition Problem to Determine Beliefs

- Simplifying assumption: HHs can only acquire **Gaussian** signals about **current** r
  - $\Rightarrow$ Prior beliefs in each period can be summarized by: $r \sim N(\mu, \Sigma)$

Information Acquisition Problem to Determine Beliefs

- Simplifying assumption: HHs can only acquire **Gaussian** signals about **current** r
- Households choose signal variance Σ_e, anticipating choices of $\mathbf{c}$ and $\mathbf{d}'$:

$$V(\mathbf{x}) = \max_{\Sigma_e} \mathbf{E}\left[U(\mathbf{c}, m(\mathbf{d}')) + \beta V\left(\mathbf{x}'\right) \mid \mathbf{x}\right]$$

Information Acquisition Problem to Determine Beliefs

- Simplifying assumption: HHs can only acquire **Gaussian** signals about **current** r
- Households choose signal variance Σ_e, anticipating choices of $\mathbf{c}$ and $\mathbf{d}'$:

$$V(\mathbf{x}) = \max_{\Sigma_e} \mathbf{E}\left[U(\mathbf{c}, m(\mathbf{d}')) + \beta V\left(\mathbf{x}'\right) \mid \mathbf{x}\right] + \omega \cdot \underbrace{\log\left(1 - G\right)}_{\text{mutual info.}}$$

$$G = \frac{\Sigma}{\Sigma + \Sigma_e}$$

$$\widehat{\mathbf{E}}(r) = (1 - G)\mu + G\left(r + e\right), \quad e \sim N(0, \Sigma_e)$$

Information Acquisition Problem to Determine Beliefs

- Simplifying assumption: HHs can only acquire **Gaussian** signals about **current** r
- Households choose signal variance Σ_e, anticipating choices of $\mathbf{c}$ and $\mathbf{d}'$:

$$V(\mathbf{x}) = \max_{\Sigma_e} \mathbf{E}\Big[U(\mathbf{c}, m(\mathbf{d}')) + \beta V\left(\mathbf{x}'\right) \mid \mathbf{x}\Big] + \omega \cdot \underbrace{\log\left(1 - G\right)}_{\text{mutual info.}}$$

$$G = \frac{\Sigma}{\Sigma + \Sigma_e}$$

$$\widehat{\mathbf{E}}(r) = (1 - G)\mu + G\left(r + e\right), \quad e \sim N(0, \Sigma_e)$$

$$\mu' = (1 - \rho)\bar{r} + \rho\widehat{\mathbf{E}}(r), \quad \Sigma' = \rho^2\Sigma(1 - G) + \sigma^2$$

Information Acquisition Problem to Determine Beliefs

- Simplifying assumption: HHs can only acquire **Gaussian** signals about **current** r
- Households choose signal variance Σ_e, anticipating choices of $\mathbf{c}$ and $\mathbf{d}'$:

$$V(\mathbf{x}) = \max_{\Sigma_e} \mathbf{E}\Big[U(\mathbf{c}, m(\mathbf{d}')) + \beta V(\mathbf{x}') \mid \mathbf{x}\Big] + \omega \cdot \underbrace{\log(1-G)}_{\text{mutual info.}}$$

$$G = \frac{\Sigma}{\Sigma + \Sigma_e}$$

$$\widehat{\mathbf{E}}(r) = (1-G)\mu + G(r+e), \quad e \sim N(0, \Sigma_e)$$

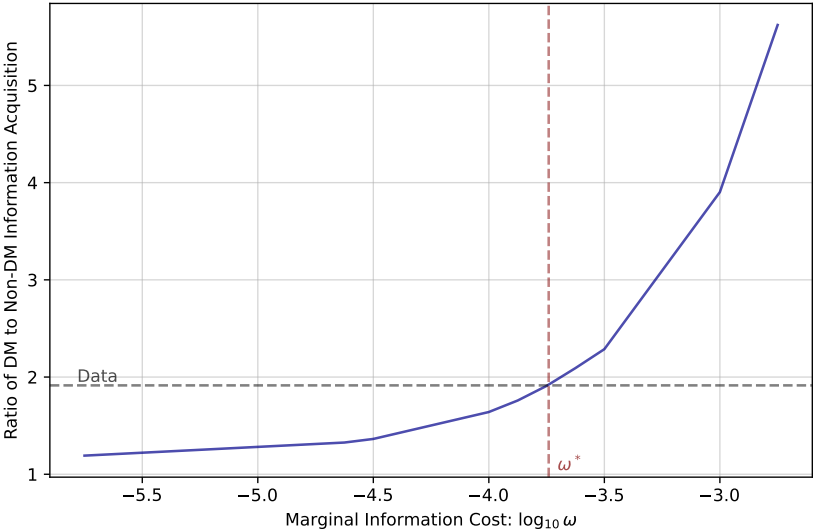
$$\mu' = (1-\rho)\bar{r} + \rho\widehat{\mathbf{E}}(r), \quad \Sigma' = \rho^2\Sigma(1-G) + \sigma^2$$

- $\mathbf{c}, \mathbf{d}'$ maximize objective with **belief errors**
  - Lower $\Sigma_e \Rightarrow \widehat{\mathbf{E}}(r) \longrightarrow r \Rightarrow$ **errors** $\longrightarrow 0 \Rightarrow$ smaller utility loss from $\mathbf{c}, \mathbf{d}'$

Calibrated Parameters

Parameter	Description	Value	Source
Internally-Calibrated			
β	Discount factor	0.9829	Asset-to-GDP ratio
ψ	Non-durables exponent	0.627	Durable-to-nondurable consumption ratio
f	Fixed cost	0.11	Adjustment probability
$1-\overline{\xi}$	Match-quality shock probability	0.034	Share of adjustments from MQ shocks
ω	Marginal information cost	$10^{-3.741}$	Concentration in information acquisition
Externally-Calibrated			
γ	RRA (and inverse EIS)	2	·
ε	Durables elasticity of substitution	0.5	McKay and Wieland (2021)
$1-\lambda$	Required downpayment	0.2	McKay and Wieland (2021)
δ	Depreciation rate	0.017	McKay and Wieland (2021)
χ	Maintenance share	0.35	McKay and Wieland (2021)
ν	Operating cost	0.012	McKay and Wieland (2021)
ρ_y	Income persistence	0.977	Flodén and Lindé (2001)
σ_ϵ	Income shock std. dev.	0.058	Flodén and Lindé (2001)
$\overline{r}$	Real rate mean	0.0143	10-Year Treasury Rate: 1961-2024
ρ	Real rate persistence	0.979	10-Year Treasury Rate: 1961-2024
σ	Real rate shock std. dev.	0.0014	10-Year Treasury Rate: 1961-2024
τ_b	Borrowing spread	0.4156%	30-Year Fixed Mortgage Rate: 1971-2024

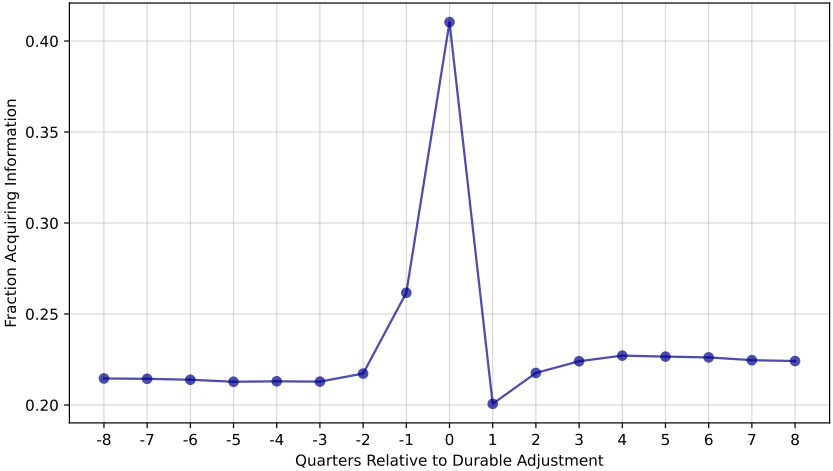
Effect of Information Cost on Information Acquisition



Summary Stats
Durables Adjustment Hazard

Selective Inattention at the Micro-Level

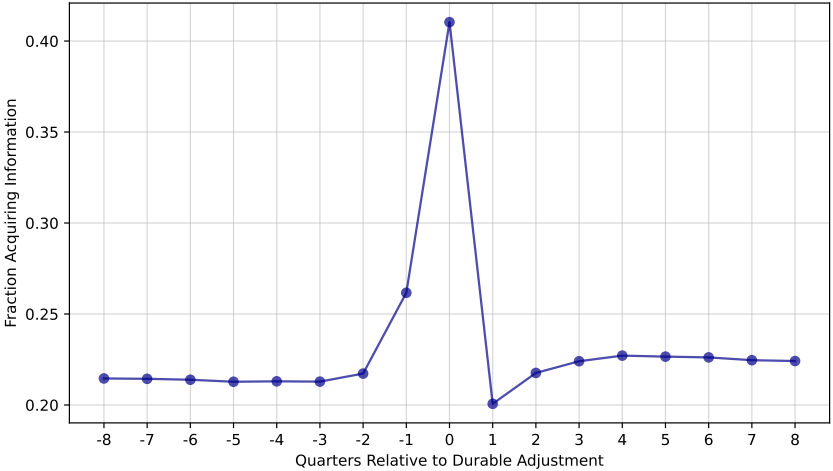
Extensive Margin of Info. Acquisition in Event-Time



Households acquire information in **all** periods...

Durables Share
Information Acquisition sS

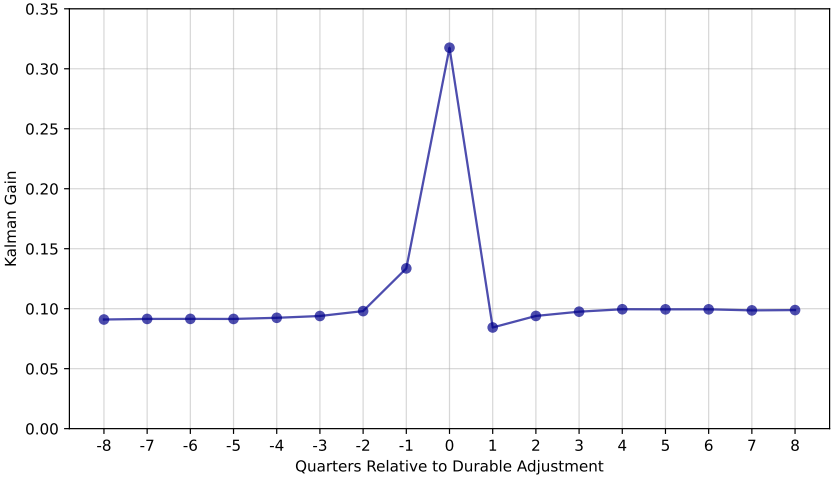
Extensive Margin of Info. Acquisition in Event-Time



... but this information acquisition is **concentrated** around durables adjustments

Durables Share
Information Acquisition sS

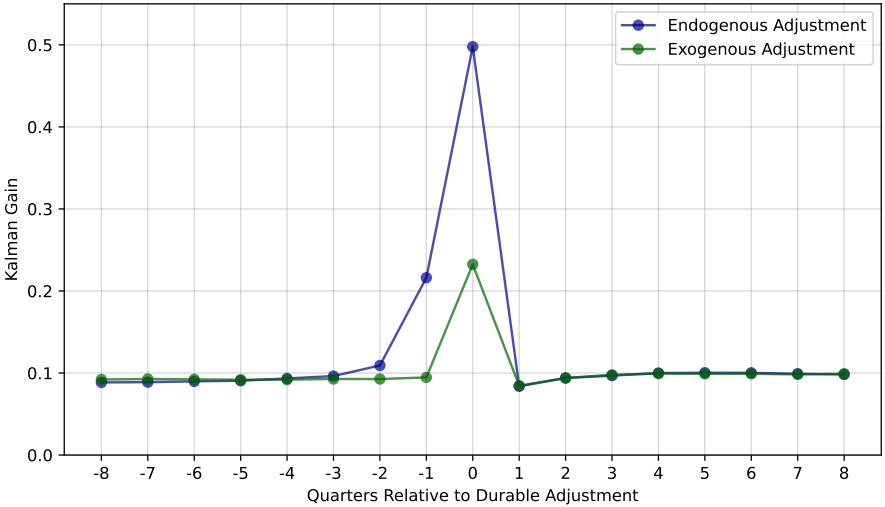
Intensive Margin of Info. Acquisition in Event-Time



Increase in information acquisition is even larger on **intensive** margin

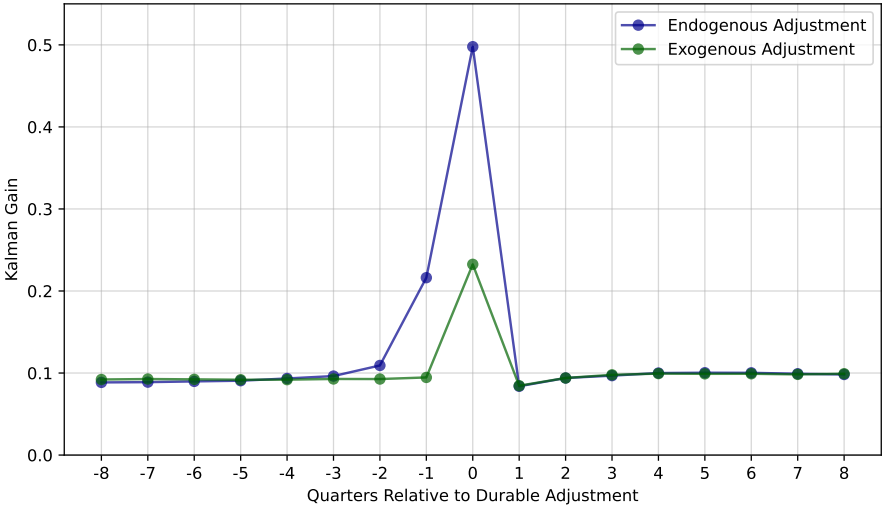
Durables Share
Information Acquisition sS

Endogenous Adjustments ⇒ Info. Acquisition Pre-Choice



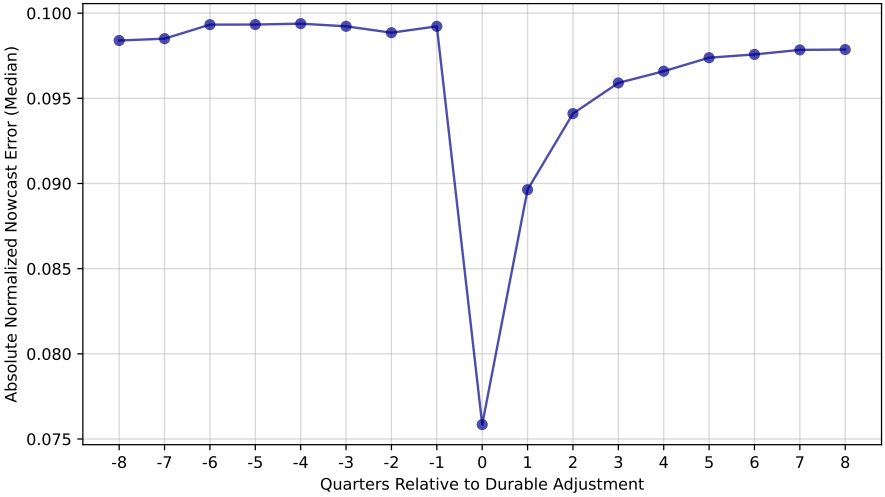
Information acquisition increases in anticipation of **state-dependent** adjustments...

Endogenous Adjustments ⇒ Info. Acquisition Pre-Choice



... but is concentrated around the choice for **time-dependent** adjustments Afrouzi et al. 24

Nowcast Errors in Event-Time



Nowcast errors remain lower **post-choice** because beliefs are a “stock” not “flow”

Implications for Aggregate Beliefs

Aggregate Beliefs are Sluggish, Like in the Data...

- Direct evidence of information-rigidity = CG (2015) regression

$$\underbrace{r_{t+3} - \overline{F}_t r_{t+3}}_{\text{forecast error}} = \alpha + \beta_{CG} \underbrace{\left(\overline{F}_t r_{t+3} - \overline{F}_{t-1} r_{t+3}\right)}_{\text{forecast revision}} + \epsilon_t$$

- Common finding: $\beta_{CG} > 0 \Rightarrow$ aggregate expectations are **sluggish**

- In a sticky-information model (constant probability of updating expectations),

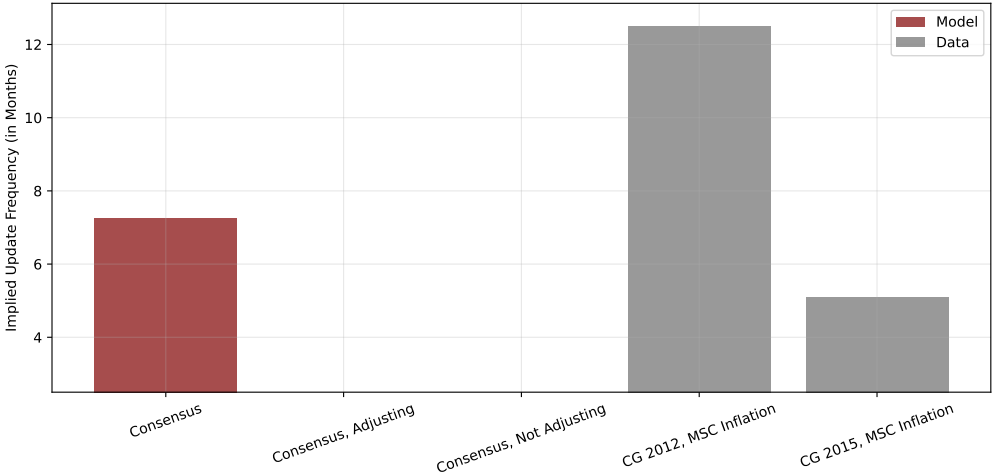
$$\text{Implied Update Frequency} = 3\left(1 + \beta_{CG}\right) \text{ Months}$$

$\Rightarrow$ Common target for calibrating sticky information models (e.g. McKay-Wieland 2021)

▸ Welfare Loss

Aggregate Beliefs are Sluggish, Like in the Data...

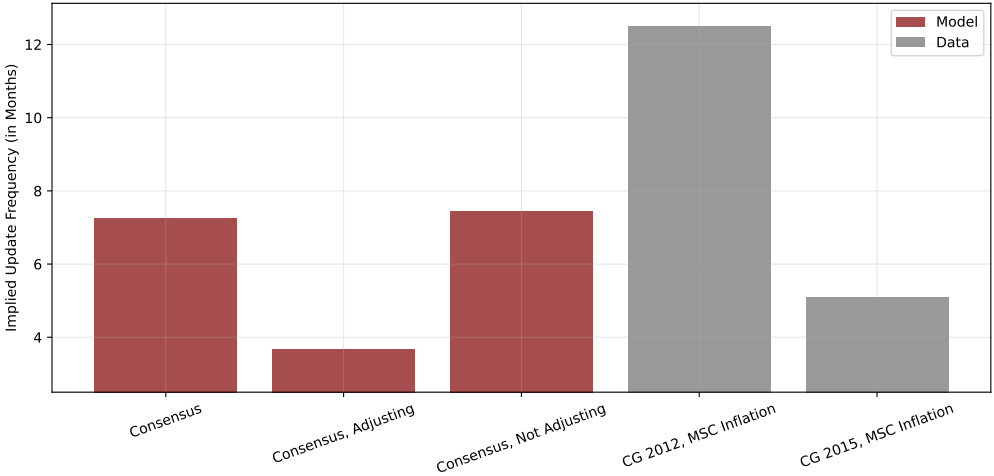
Implied Update Frequency $= 3\,(1 + \beta_{CG})$ Months



Welfare Loss

… But This Masks Substantial Selection into Attention!

Implied Update Frequency $= 3\,(1 + \beta_{CG})$ Months



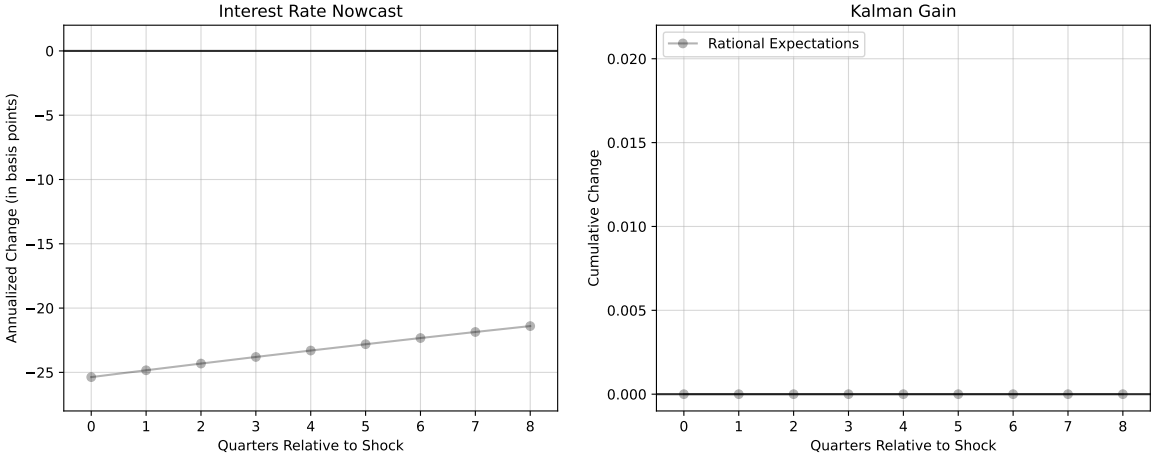
Welfare Loss

Outline

1. Existing Surveys: Expectations Accuracy around Decision-Making
2. New Survey: Information Acquisition around Decision-Making
3. Incomplete Markets Model with Selective Inattention
4. Interest Rate Passthrough with Selective Inattention
5. Conclusion

Responses to Interest Rate Cuts

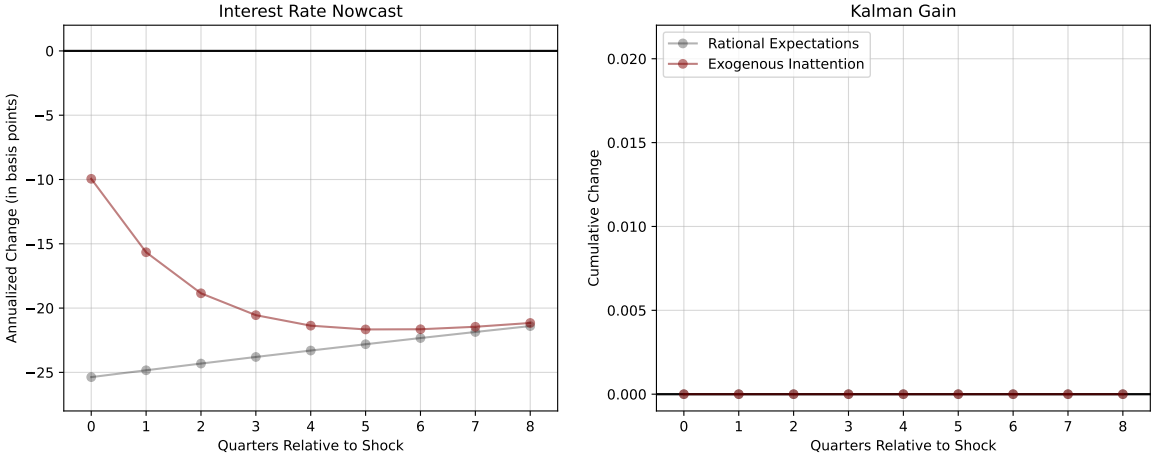
Impulse Response of Beliefs to Rate Cut



Rational expectations: households observe current interest rate

▸ Decomposition

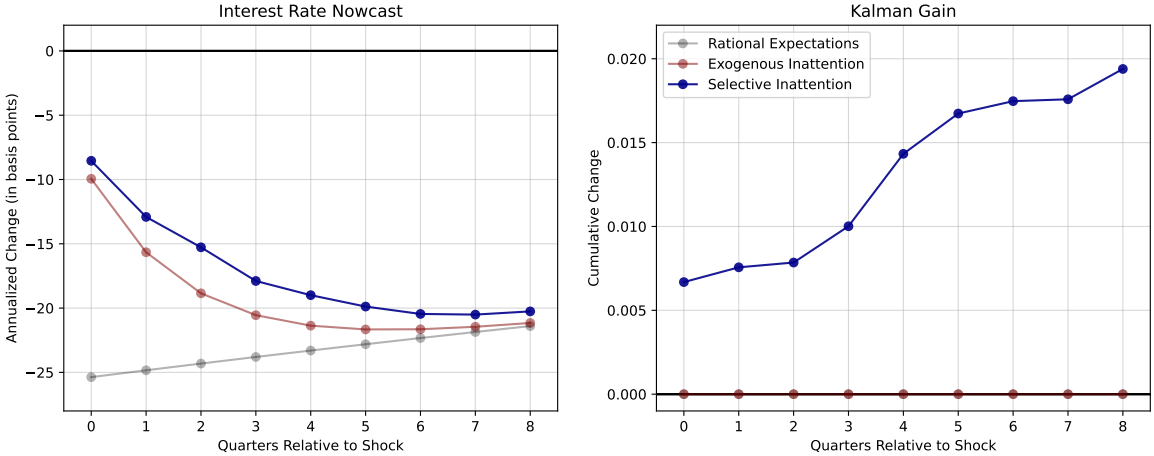
Impulse Response of Beliefs to Rate Cut



Exogenous inattention: constant G set to match CG 15 coefficient in baseline model

Decomposition

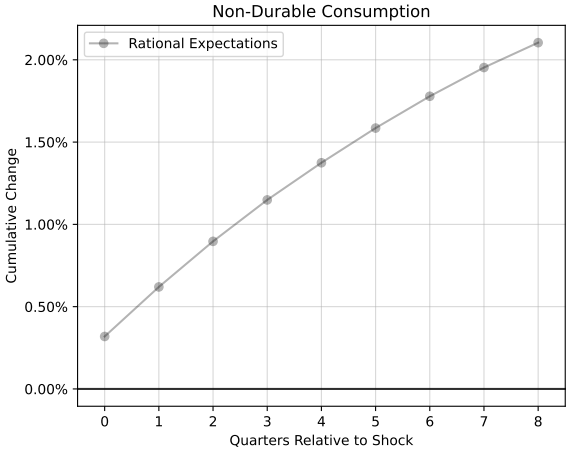
Impulse Response of Beliefs to Rate Cut

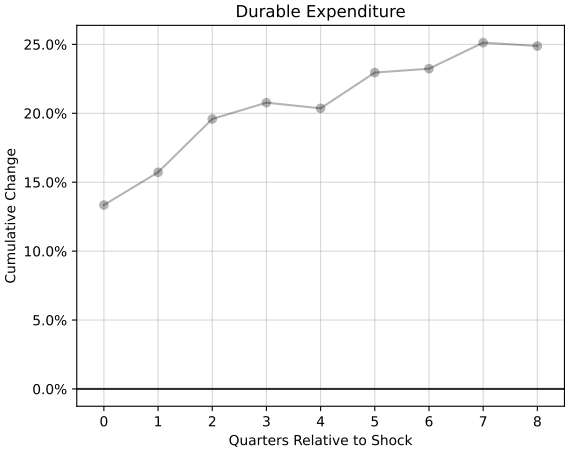


Selective inattention: baseline model with endogenous information acquisition

Decomposition

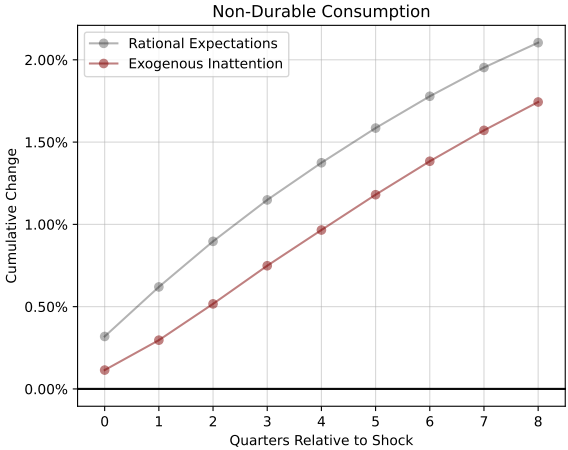
Impulse Response of Spending to Rate Cut

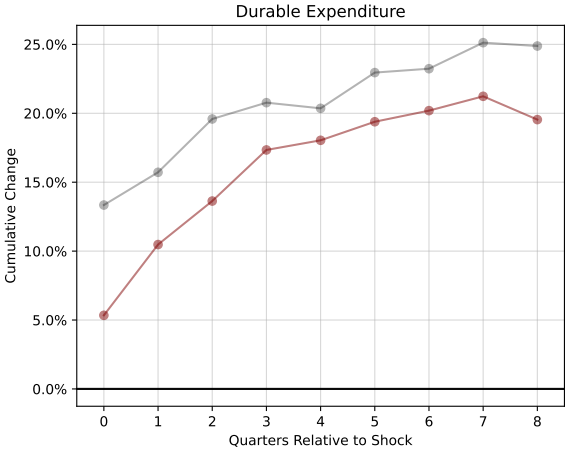




Incorporating GE Effects
Non-Linearity

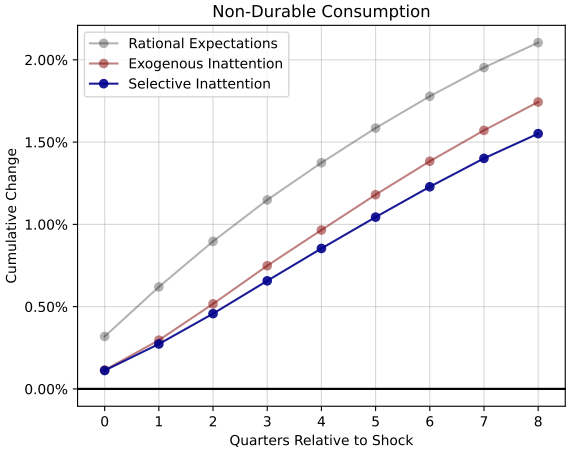
Impulse Response of Spending to Rate Cut

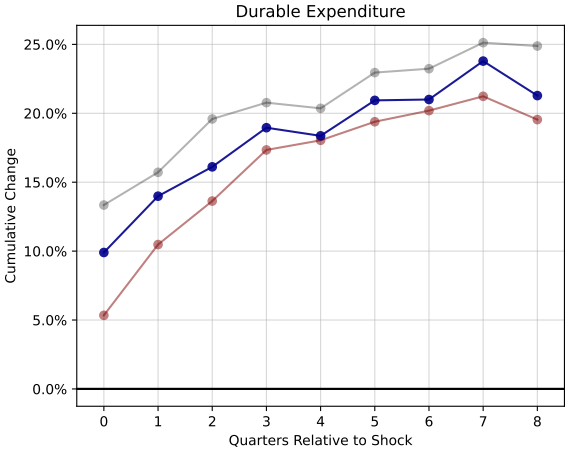




Incorporating GE Effects
Non-Linearity

Impulse Response of Spending to Rate Cut

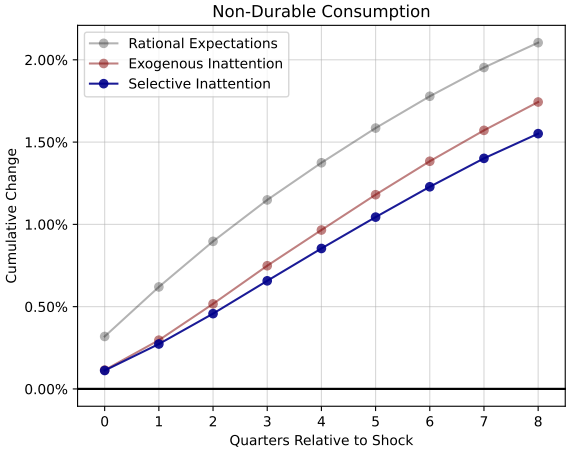


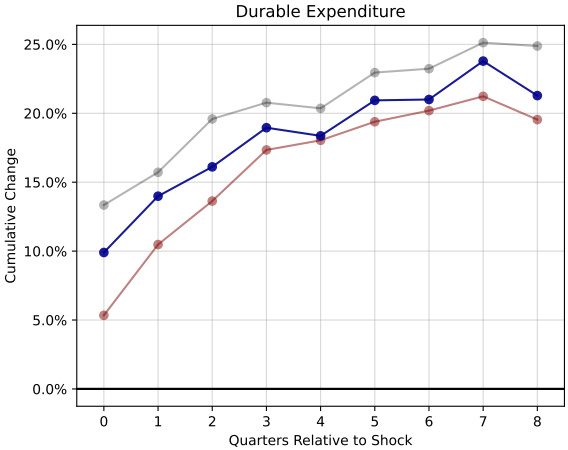


Non-durable response is smaller relative to **exogenous inattention**...

▸ Incorporating GE Effects ▸ Non-Linearity

Impulse Response of Spending to Rate Cut

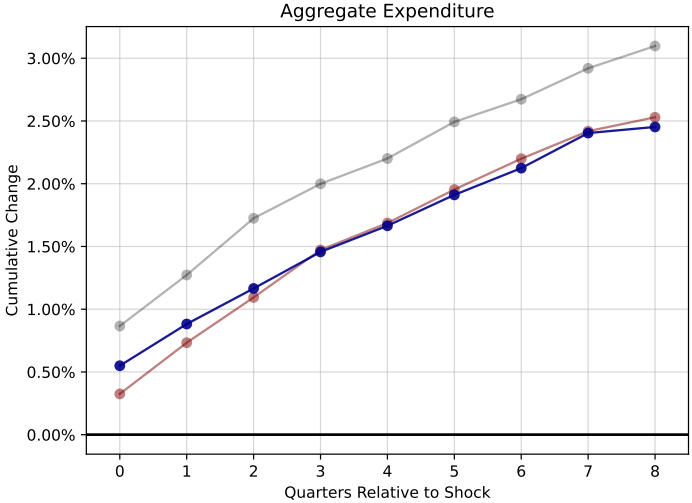




... but **durable** response is larger, closer to **rational expectations**!

Incorporating GE Effects
Non-Linearity

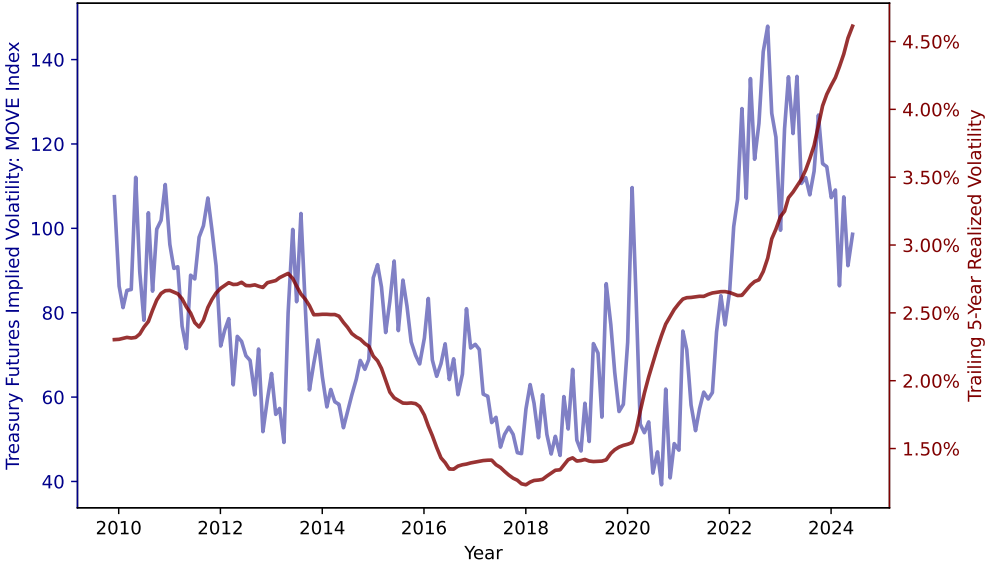
Small Difference in Aggregate Spending Response to Rate Cut



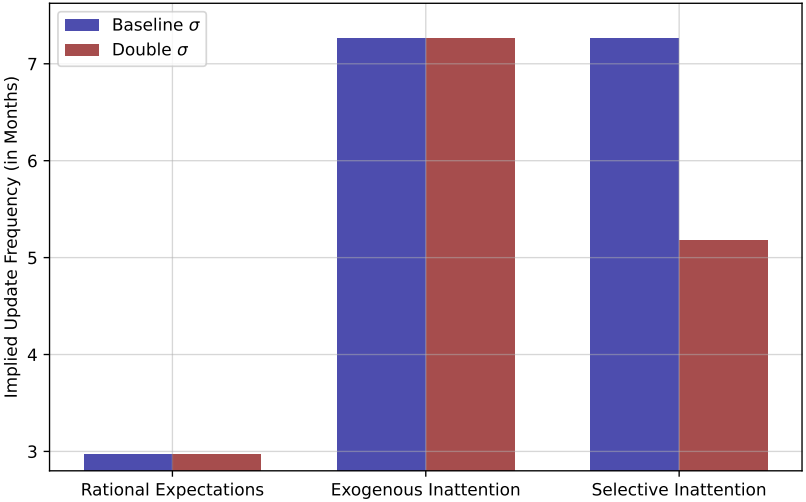
Shift in **composition** of spending to durables $\Rightarrow$ GE effects through $\frac{p_d}{p_c}$ Barsky et al. 07

Effects of Changes in Interest Rate Volatility

Motivation: Recent Rise in Rate Volatility

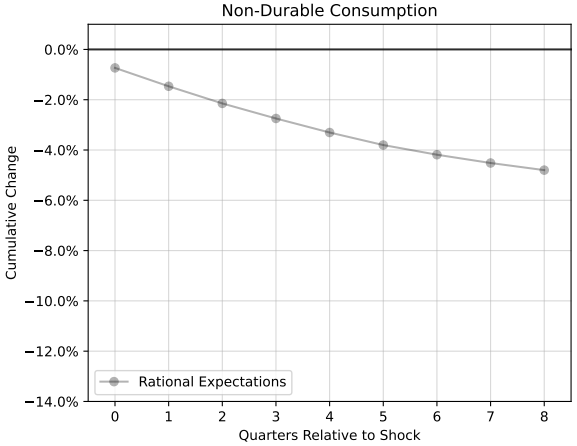


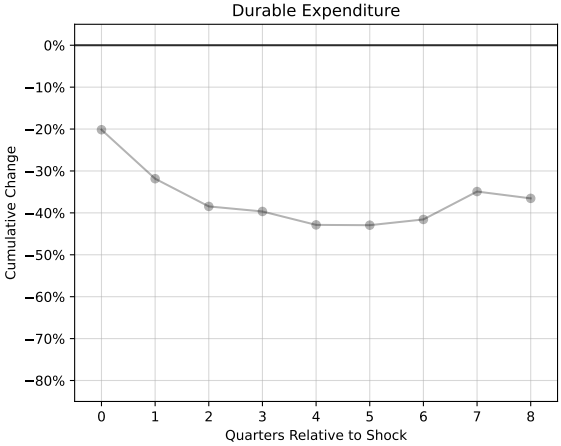
Effects of Increase in Volatility on Beliefs



Increase in volatility $\Rightarrow$ more information acquisition $\Rightarrow$ **less** belief rigidity

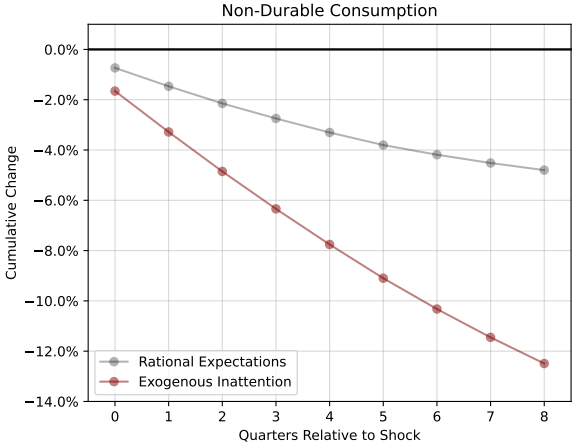
Impulse Response of Spending to Increase in Volatility

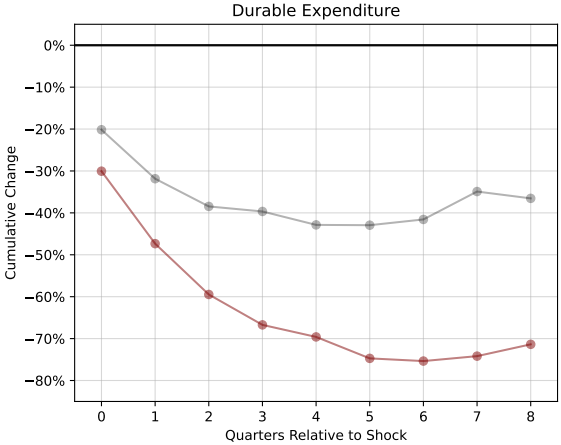




Increase in volatility ⇒ spending falls due to precautionary motives... Sandmo 70

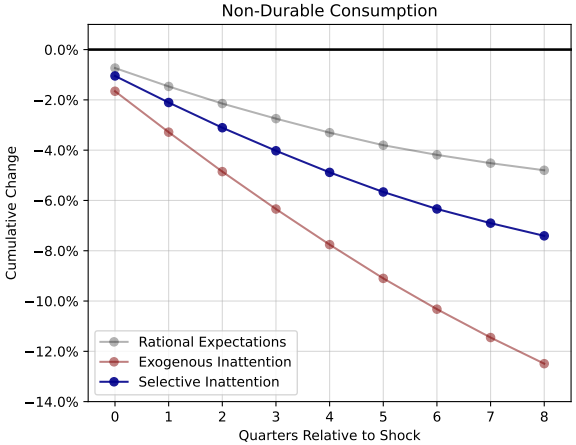
Impulse Response of Spending to Increase in Volatility

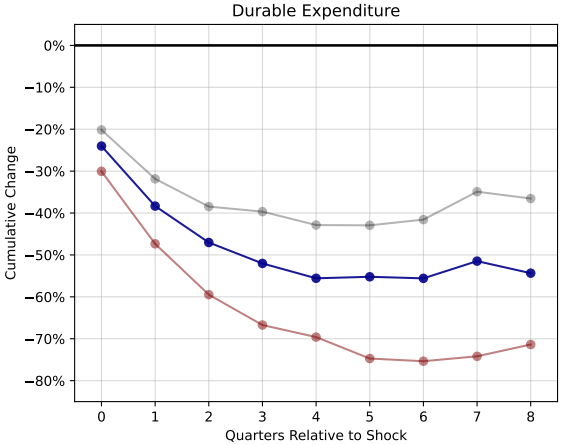




... which is stronger with **exogenous inattention** because of additional uncertainty

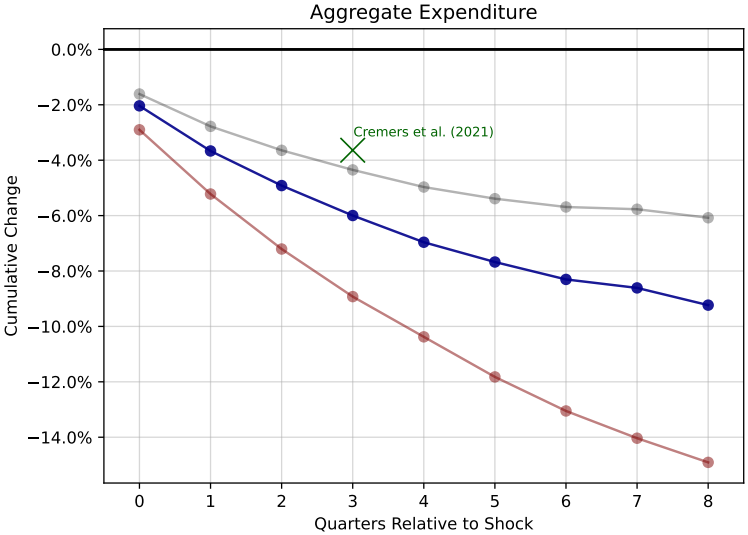
Impulse Response of Spending to Increase in Volatility





Selective inattention undoes $\sim$ 60% of this fall due to $\uparrow$ info. acquisition!

Response of Aggregate Spending is Closer to the Data



State-Dependence of Rate Cuts

Outline

1. Existing Surveys: Expectations Accuracy around Decision-Making
2. New Survey: Information Acquisition around Decision-Making
3. Incomplete Markets Model with Selective Inattention
4. Interest Rate Passthrough with Selective Inattention
5. Conclusion

Conclusion

- Data: Households are selectively inattentive to interest rates
  - IA is concentrated around durables purchases, where beliefs are more accurate
- Model: **Like** exogenous inattention, selective inattention generates:
  1. Slow-moving aggregate beliefs
  2. Dampened responses of non-durable consumption to interest rates

Conclusion

- Data: Households are selectively inattentive to interest rates
  - IA is concentrated around durables purchases, where beliefs are more accurate
- Model: **Like** exogenous inattention, selective inattention generates:
  1. Slow-moving aggregate beliefs
  2. Dampened responses of non-durable consumption to interest rates
- Model: **Unlike** exogenous inattention, selective inattention:
  1. Shifts the **composition** of spending responses to rate cuts towards durables
  2. **Dampens** responses to changes in volatility closer to the data
  3. **Accelerates** the impact of larger rate cuts
  4. **Reduces** state dependence of rate cuts on volatility

Conclusion

- Data: Households are selectively inattentive to interest rates
  - IA is concentrated around durables purchases, where beliefs are more accurate
- Model: **Like** exogenous inattention, selective inattention generates:
  1. Slow-moving aggregate beliefs
  2. Dampened responses of non-durable consumption to interest rates
- Model: **Unlike** exogenous inattention, selective inattention:
  1. Shifts the **composition** of spending responses to rate cuts towards durables
  2. **Dampens** responses to changes in volatility closer to the data
  3. **Accelerates** the impact of larger rate cuts
  4. **Reduces** state dependence of rate cuts on volatility

Takeaway: Right model of belief formation may be different for lumpy decisions, such as durables, price setting, investment, hiring and firing, ...

THANK YOU!

tdesilva@stanford.edu

www.timdesilva.me

Main Survey Questions

- Eliciting our main measure of information acquisition

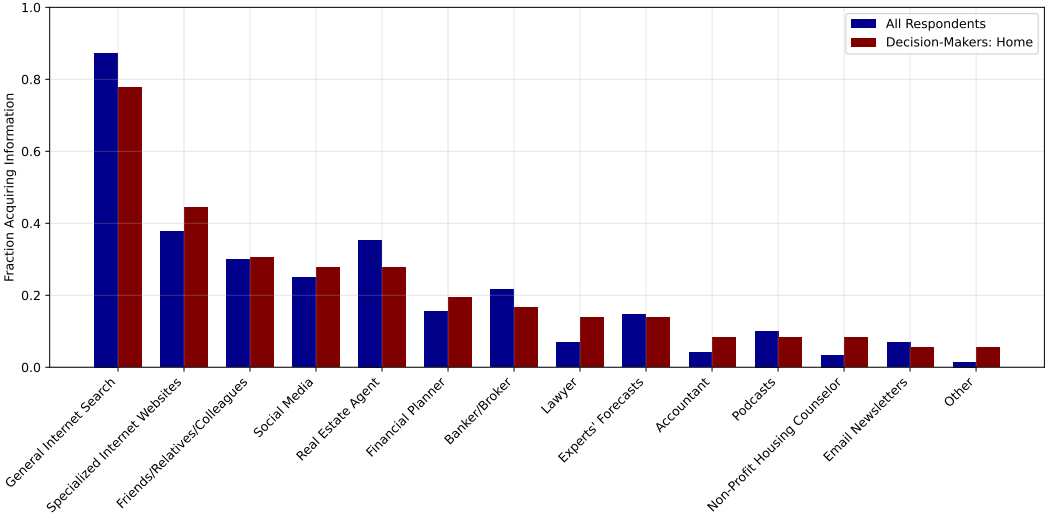
  Step 1: *In the last 3 years, did you actively search for information about any of the following economic variables in the U.S.?*
  By "active search" we mean a deliberate effort to find information which could include searching online, reading news articles or reports, talking to a financial advisor or broker, or any other intentional effort to gather information.
  Step 2: *How many months ago did you last actively search for information about mortgage rates?*

- Eliciting households' distance from the primary home purchase

  Owners: *How many months ago did you finalize the purchase of your current primary residence?*
  Renters: *How many months from now do you expect the closing on your primary residence purchase?*
  By "closing", we mean signing the final documents to officialize the purchase.

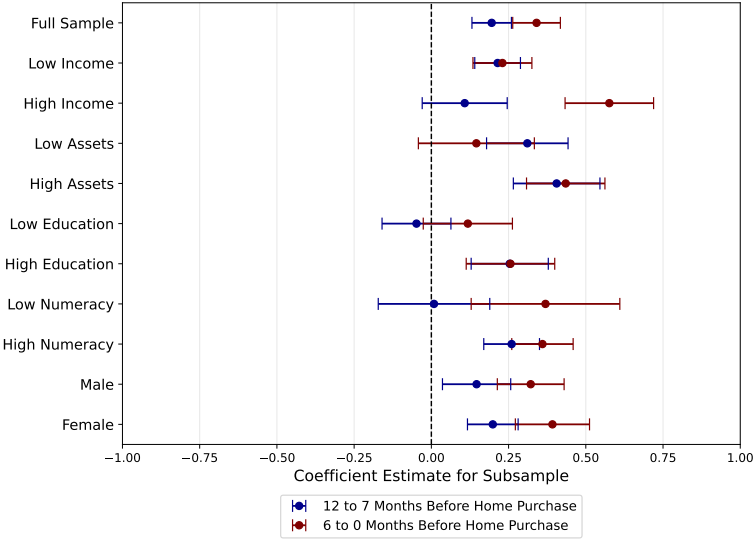
Back

Sources of Information Acquisition



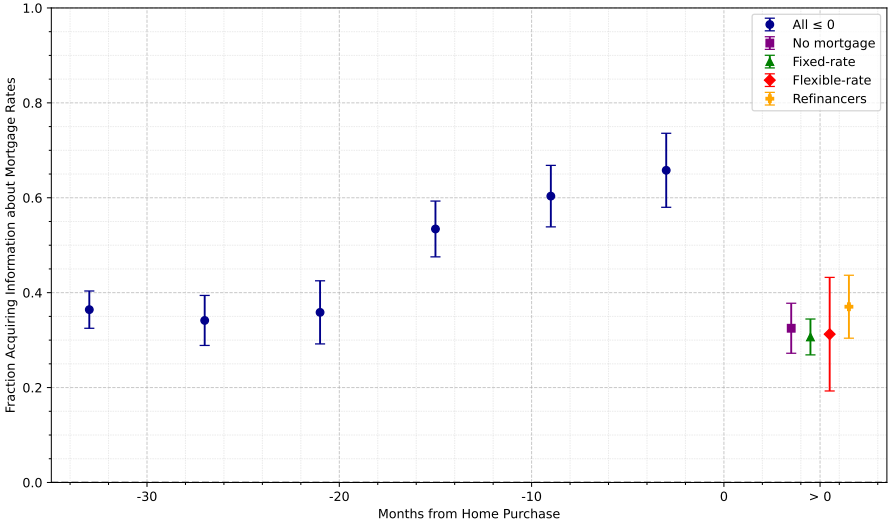
Back: IA
Back: Additional Results

Heterogeneity in Information Acquisition



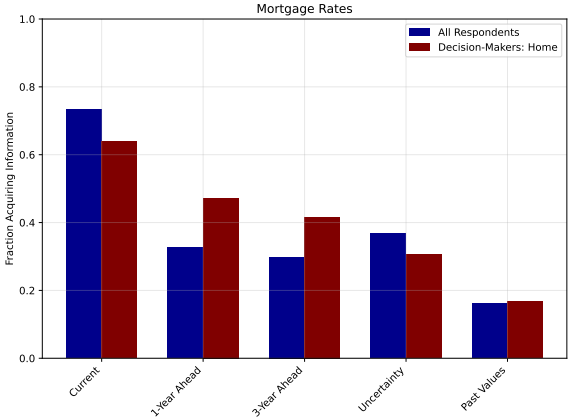
Back

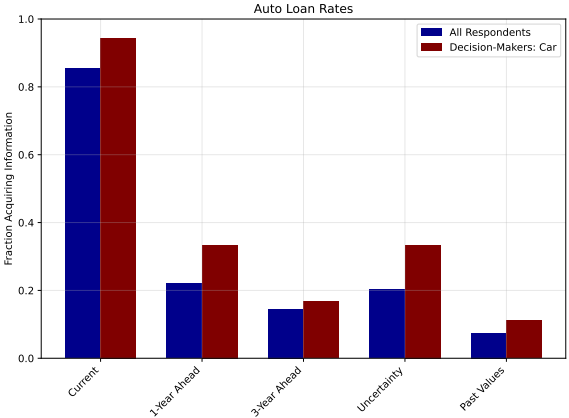
Heterogeneity in Information Acquisition of Owners



Back: IA
Back: Additional Results

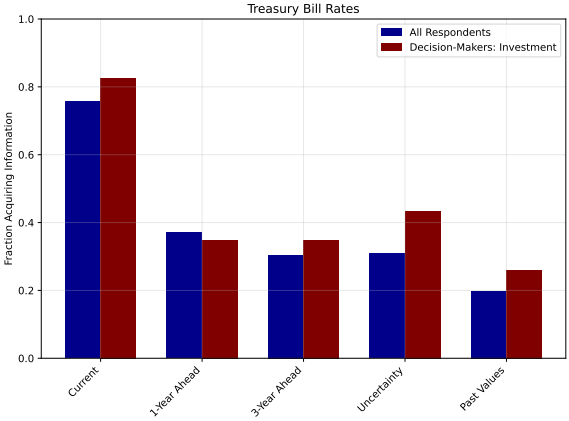
IA is Primarily about Current Values of Variables





▸ Investment Decisions ◂ Back: IA ◂ Back: Additional Results

IA is Primarily about Current Values of Variables



Back

Endogeneity of Decision-Making

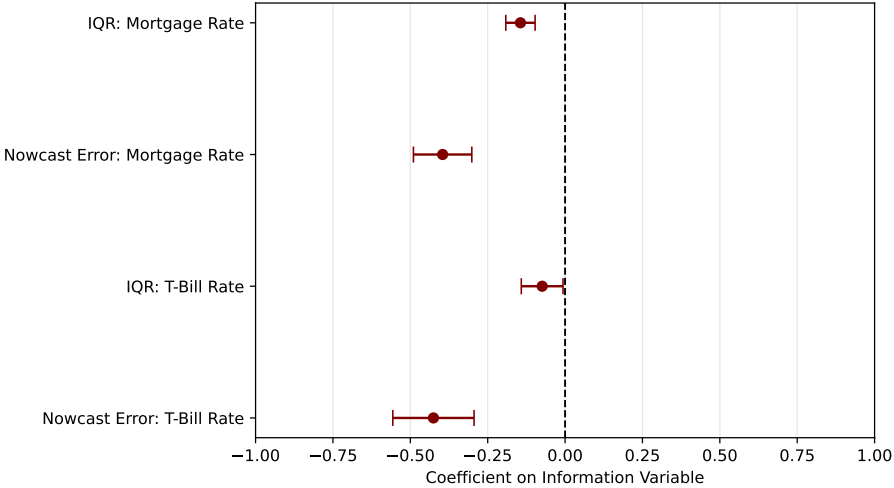
- Concern: decision-making is **endogenous** to information acquisition and beliefs
- Solution: **IV** = anticipated moves due to job relocations

Dependent Variable: Information Acquisition

Variable	OLS	First Stage	IV	OLS	First Stage	IV
Home Decision-Maker	0.30***		0.89***	0.30***		0.96***
	(0.07)		(0.34)	(0.06)		(0.34)
Job Relocation		0.23***			0.23***	
		(0.07)			(0.07)	
N	787	787	787	787	787	787
Controls				✓	✓	✓
F-stat		10.51			3.57	

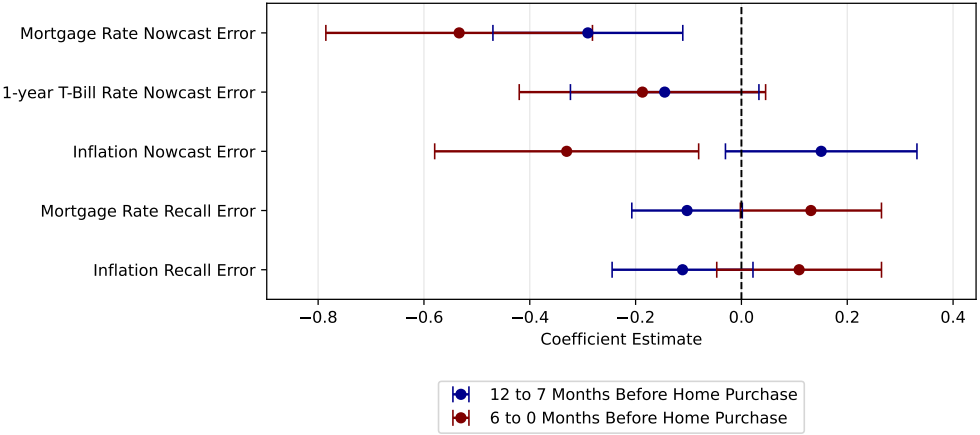
Back

Direct Association Between IA and Beliefs



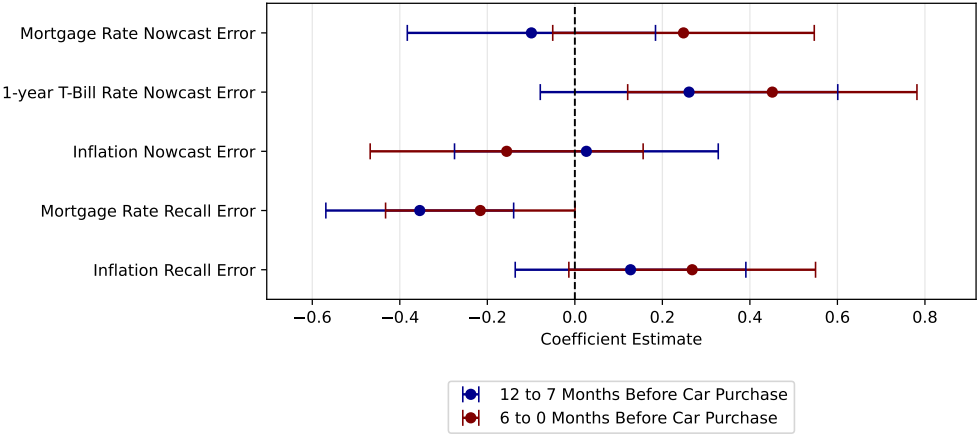
Back

Reduction in Nowcast Errors: Home Purchase



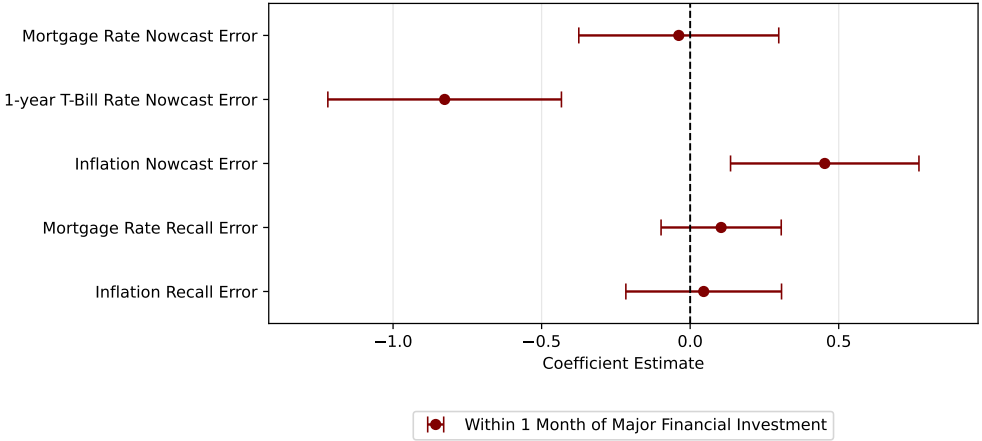
Back

Reduction in Nowcast Errors: Car Purchase



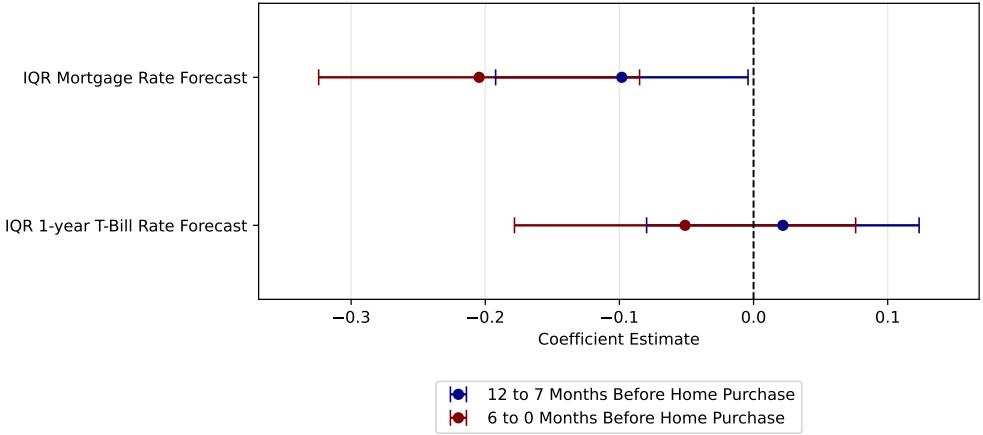
Back

Reduction in Nowcast Errors: Financial Investment



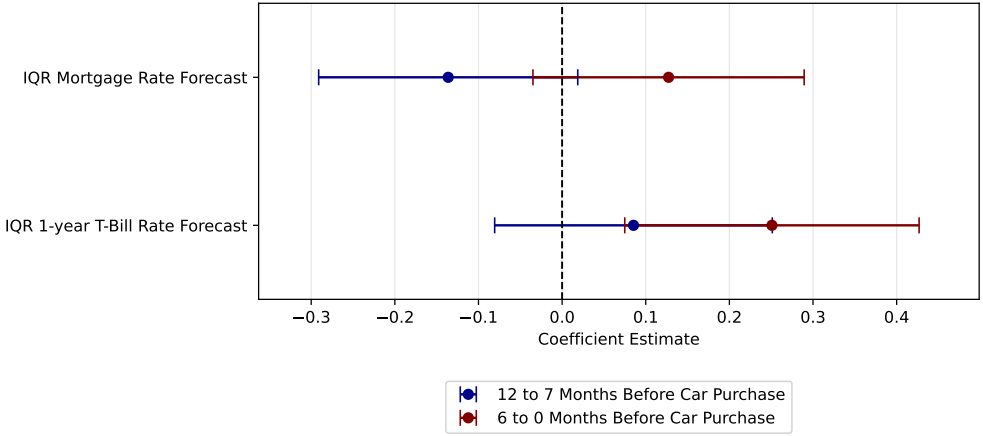
Back

Reduction in Subjective Uncertainty: Home Purchase



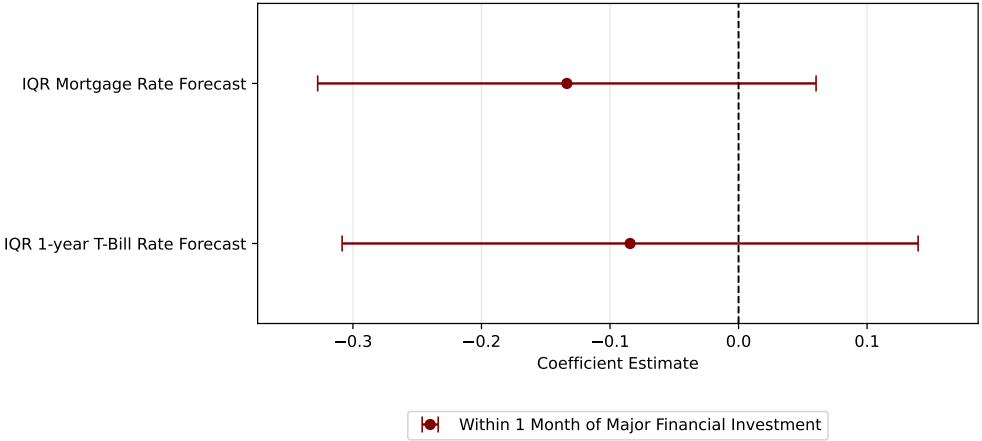
Back

Reduction in Subjective Uncertainty: Car Purchase



Back

Reduction in Subjective Uncertainty: Financial Investment



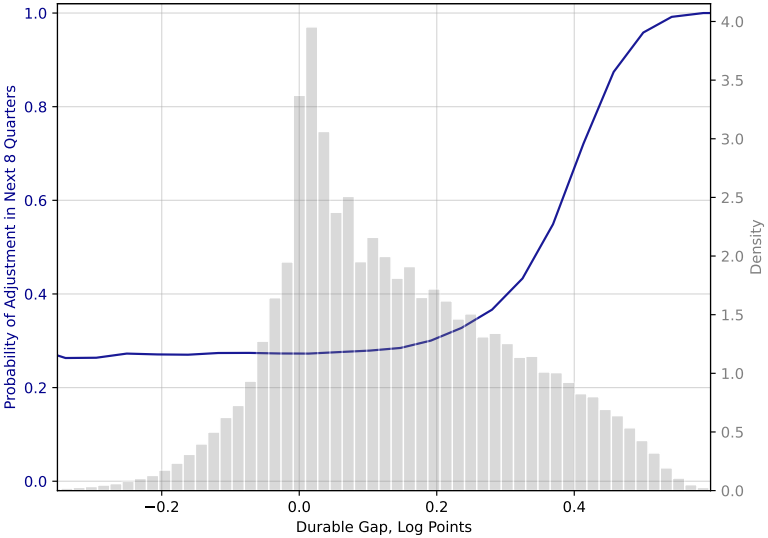
Back

Steady-State Summary Statistics

	Mean	SD	P10	P50	P90
Assets/Income: b/y	3.51	4.93	-0.91	1.91	10.27
Durable/Non-Durables: d'/c	2.55	0.40	1.99	2.58	3.01
Durables Gap	0.14	0.17	-0.05	0.11	0.38
Acquired Information	0.23	0.42	0.00	0.00	1.00
Kalman Gain: G	0.10	0.21	0.00	0.00	0.40
Kalman Gain Conditional on IA	0.46	0.21	0.30	0.40	0.80
Normalized Nowcast Error: $\lvert\hat{\mathbb{E}}(r) - r\rvert/\lvert r\rvert$	0.28	7.94	0.02	0.10	0.32
Normalized Prior Variance: Σ/σ_r^2	0.33	0.17	0.13	0.30	0.57

Back

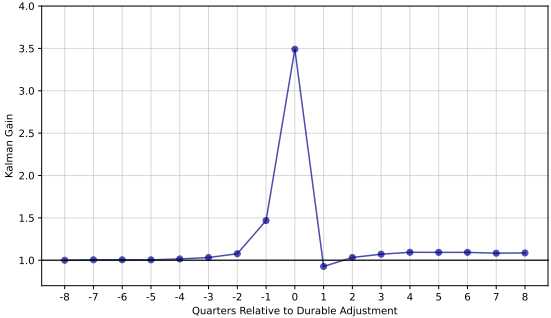
Adjustment Probability as a Function of Durables Gap



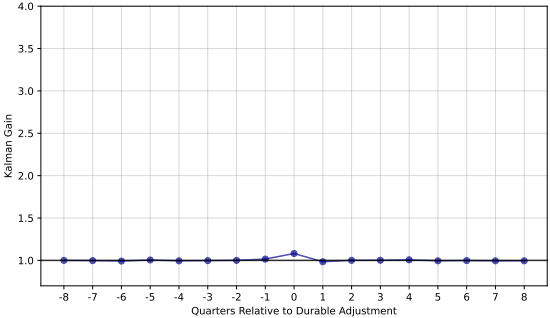
Back

Concentration in Info. Acquisition ↘ Durables Share

Baseline: $\psi = 0.63$

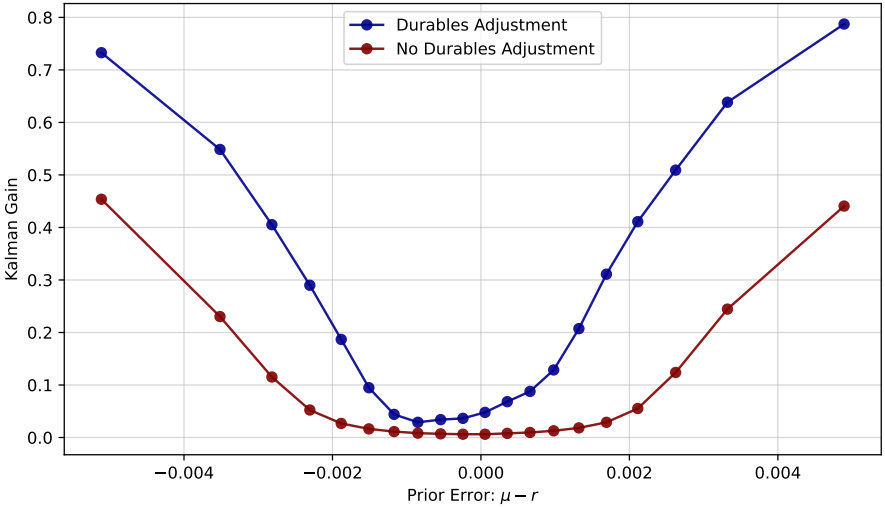


Low Durables Share: $\psi = 0.99$



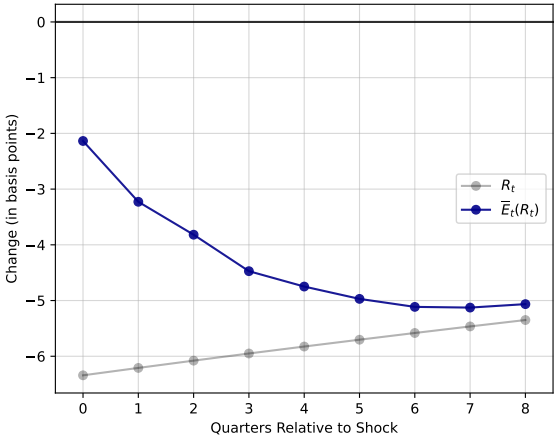
◂ Back: Extensive ◂ Back: Intensive

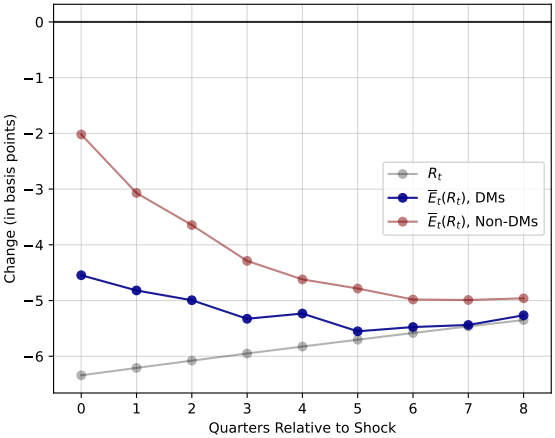
Durables Adjustment Shift sS Bands of Info. Acquisition



Back: Extensive
Back: Intensive

Decomposition of Aggregate Belief Response





Back

Welfare Loss from Inattention

- Natural question: how large are welfare losses from selective inattention?
- Compute two welfare metrics in **basis points** of lifetime consumption
  1. **Static**: loss from not having full-information in **current** period, ignoring info. cost
  2. **Dynamic**: loss from not having full-information in **all** periods, ignoring info. cost

▸ CG Coefficients

Welfare Loss from Inattention

- Natural question: how large are welfare losses from selective inattention?
- Compute two welfare metrics in **basis points** of lifetime consumption
  1. **Static**: loss from not having full-information in **current** period, ignoring info. cost
  2. **Dynamic**: loss from not having full-information in **all** periods, ignoring info. cost
- Losses are **small**, but still have aggregate effects (next)!

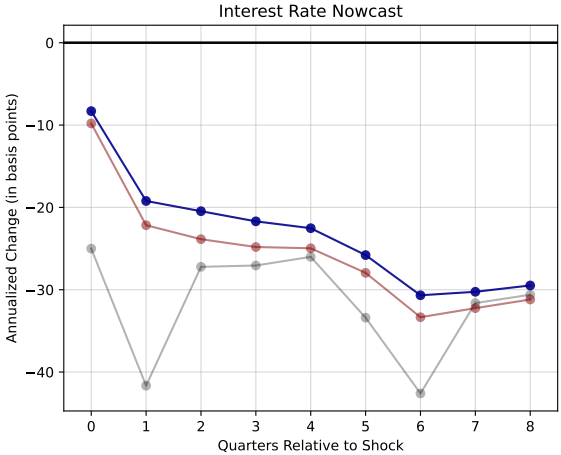
Akerlof-Yellen 85

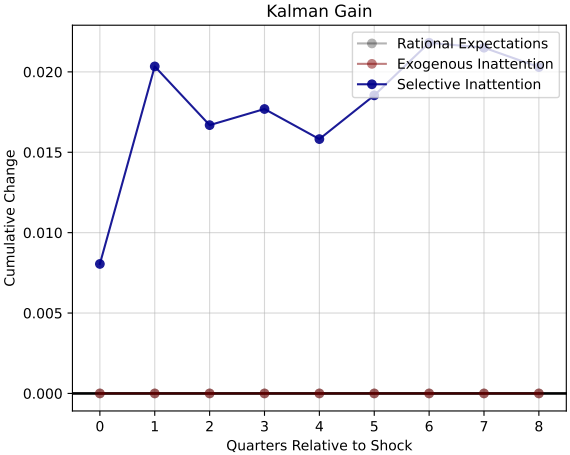
Maćkowiak-Wiederholt 15

	Static	Dynamic
Mean	0.03	2.15
Median	0.02	1.91

▸ CG Coefficients

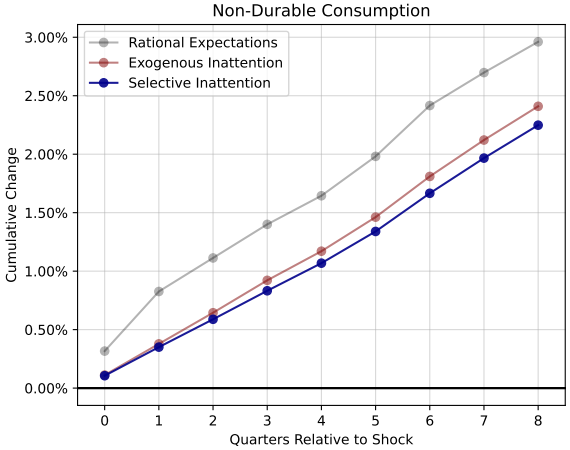
IRFs to Romer-Romer Shock with Agg. Y and P Response

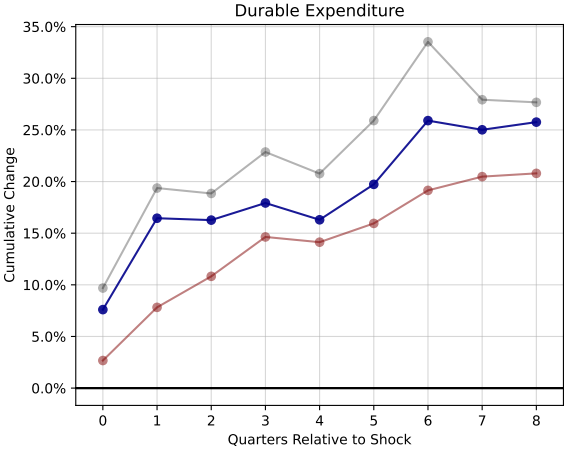




Back

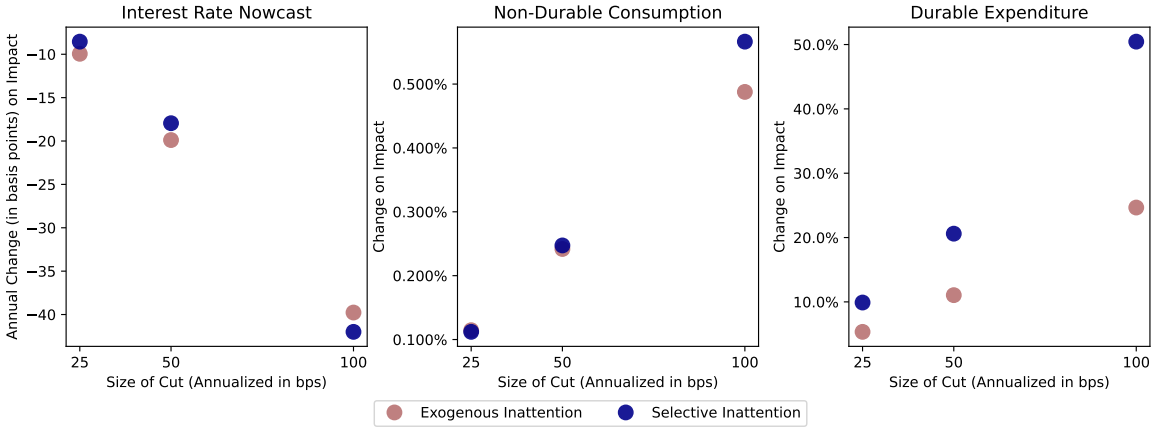
IRFs to Romer-Romer Shock with Agg. Y and P Response





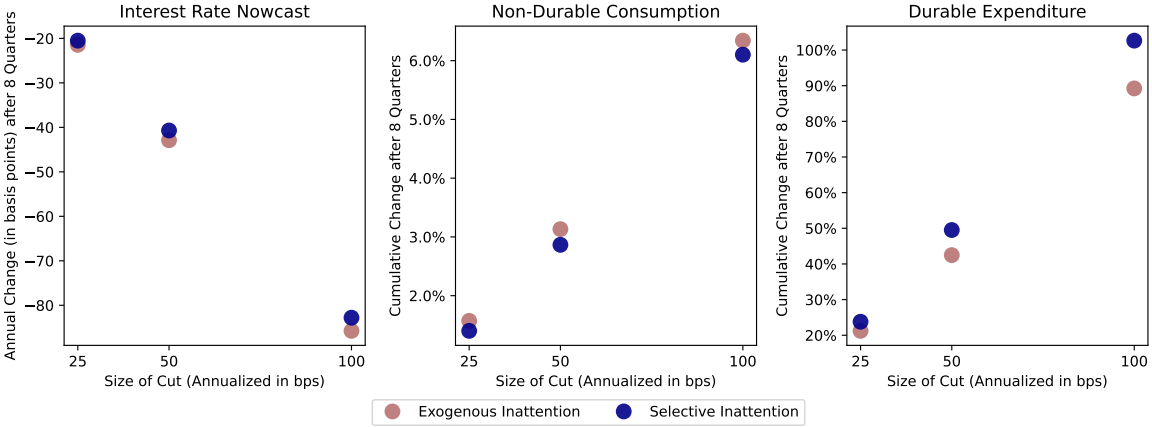
Back

Non-Linear Impact of Rate Cuts: On Impact



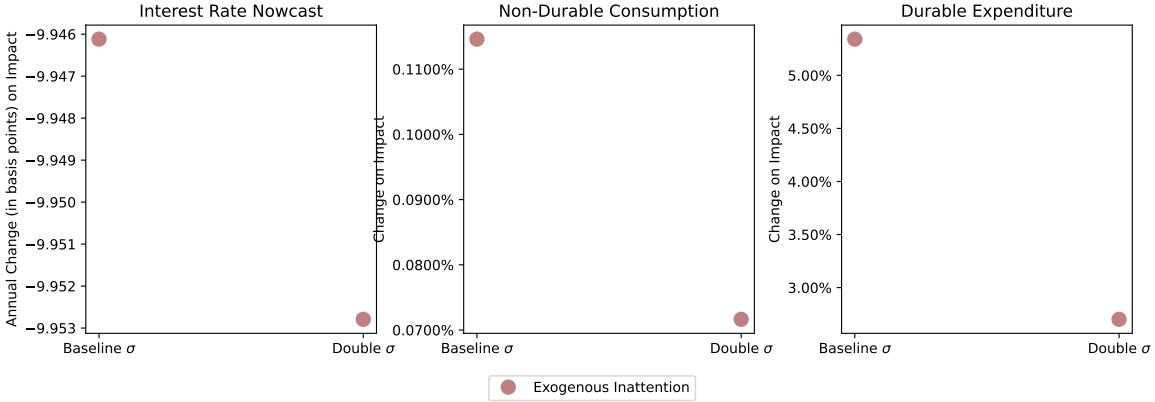
Back

Non-Linear Impact of Rate Cuts: After 8 Quarters



Back

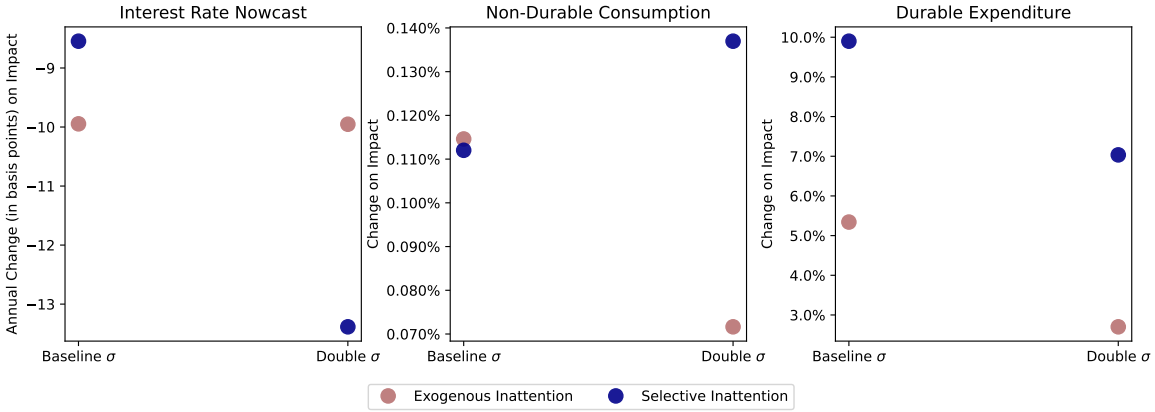
State-Dependence on Volatility: On Impact



Increase in volatility ⇒ consumption is less responsive to interest rates

Back

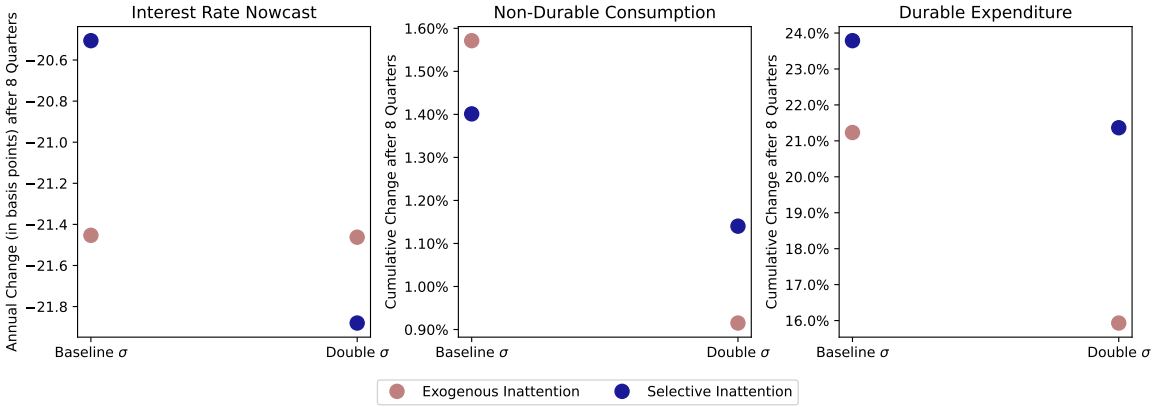
State-Dependence on Volatility: On Impact



... but not with **selective inattention** because of increased info. acquisition!

Back

State-Dependence on Volatility: After 8 Quarters



Back